Sathya Technologies, Ameerpet

Hibernate

By: Raghu



ch m

ch mm

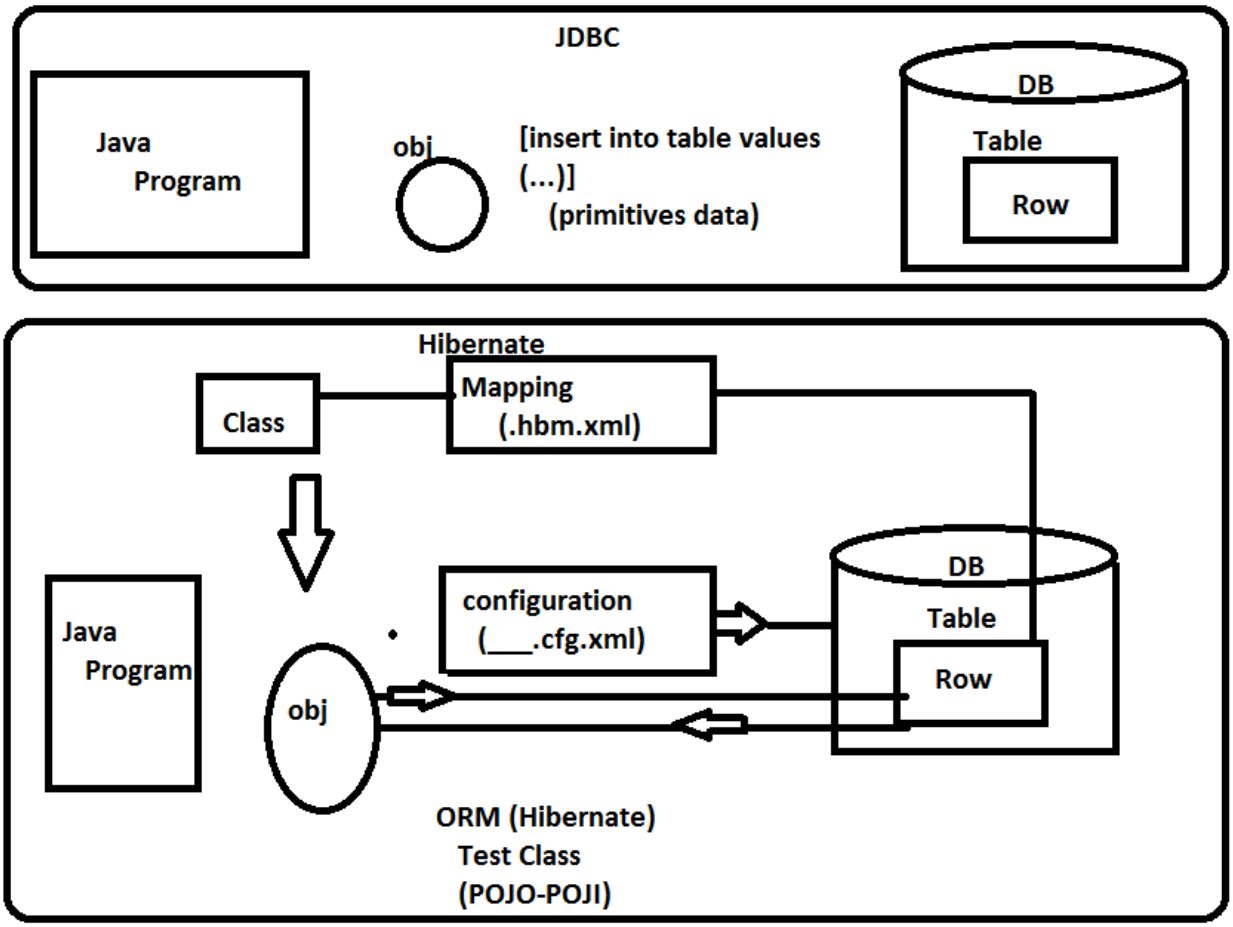
chg ch

I

javabyraghu@gmail.com






JDBC
Java
Program
obj
[insert into table values
(...)]
(primitives data)
DB
Table
Row
Hibernate
Class
Mapping
(.hbm.xml)
Java
Program
configuration
(___.cfg.xml)
obj
DB
Table
Row
ORM (Hibernate)
Test Class
(POJO-POJI)

javabyra


javabyraghu@gmail.co

javabyraghu@gmail.c

-
-

I

javabyraghu@gmail.com

➔
➔

Printing decimal number in other formats (Binary,octal,hex): -

m g g

out
out

m g

out

out

javabyraghu@gmail.com

m ch g

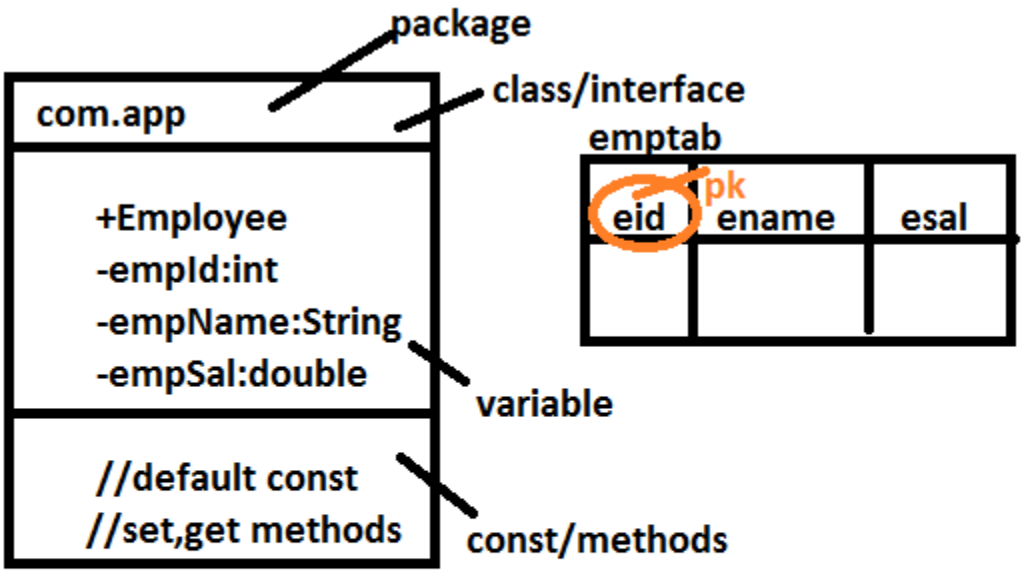
out
out
out

out
out
out

javabyraghu@gmail.com

ch g

javabyraghu@gmail.com



emptab

eid (pk)	ename	esal

+public
-private
#protected
None default

javabyraghu.com

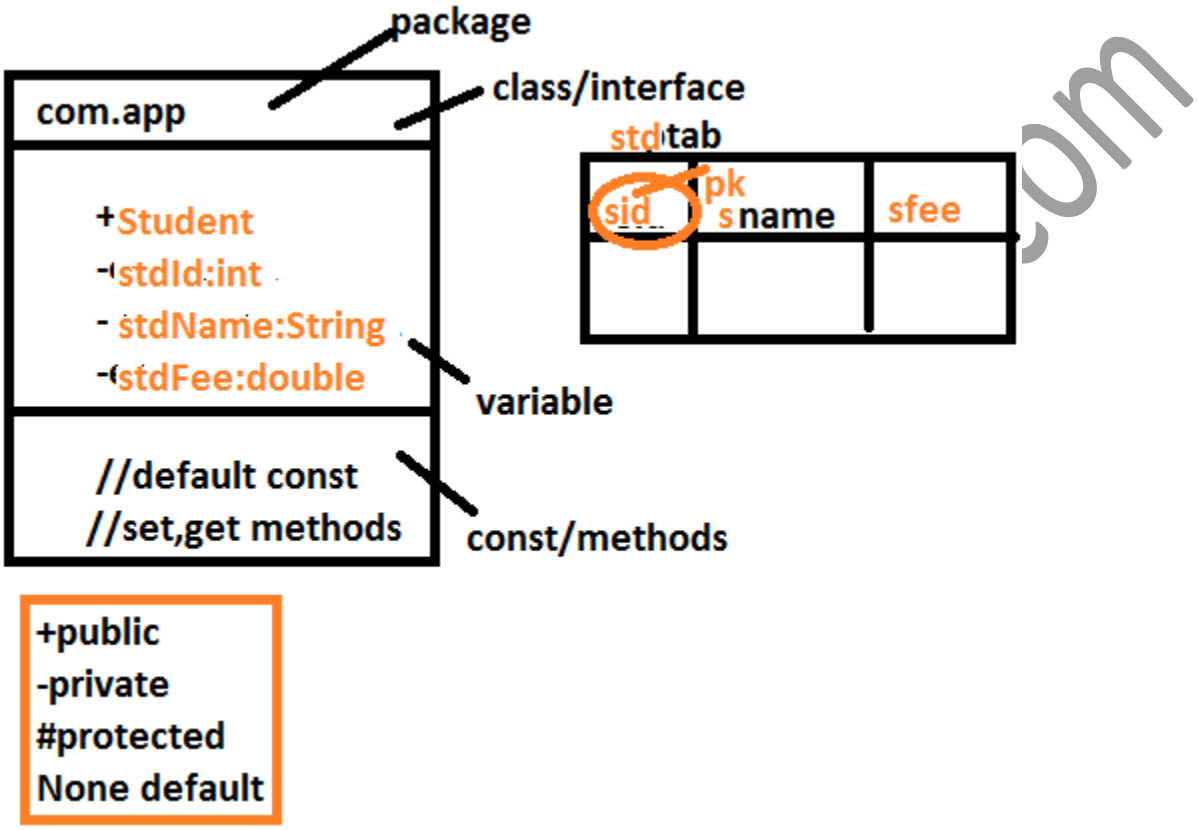
out

javabyraghu@gmail.com

gch g g ch m gm m chchch ch gchg m ch

ch

chg



+public
-private
#protected
None default

javabyr .com

out

javabyraghu@gmail.com

| |

}

@Entity: - *G g ch g ch g m*

@Id: - *G g g ch g ch*

@Table: - *m m g ch ch chg g m gm G chg m m g ch ch chg ch ch ch*

@Column: - *m m g ch m ch g G g m gm G g g m m g ch ch g ch chg m ch*

---

hibernate.cfg.xml

driver class name, url, username, password

dialect (class generates SQL).

show_sql hbm2ddl.auto

ch ch g ch

g ch ch m ch gm g ch-
g ch ch m ch gm

javabyraghu@gmail.com

g ch ch m ch gm ch ch
g ch ch m ch gm m

g ch ch g ch
g ch ch m -

g ch ch m

m ch - m g ch m ch ch

g ch ch ch

g ch ch m ch gm g ch-

g ch ch m ch gm

g ch ch m ch gm ch ch
g ch ch m ch gm m

g ch ch g ch

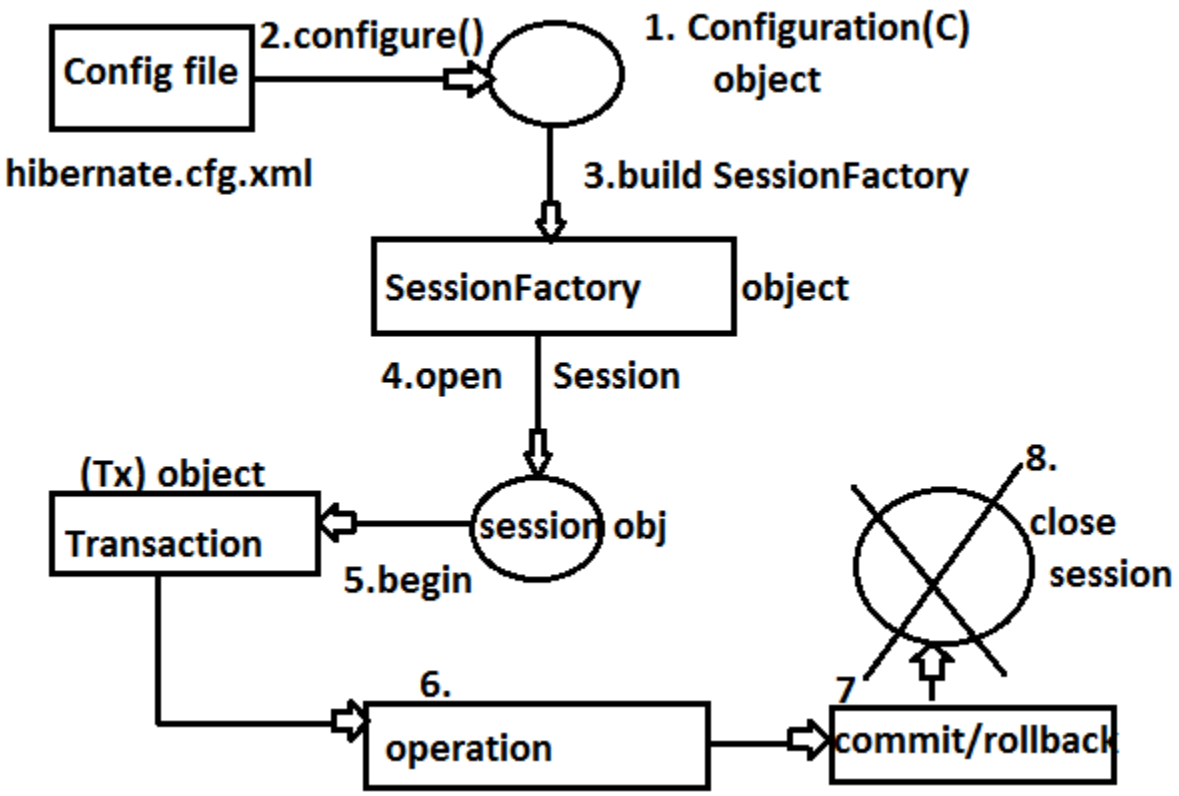
g ch ch m -

g ch ch m

javabyraghu@gmail.com

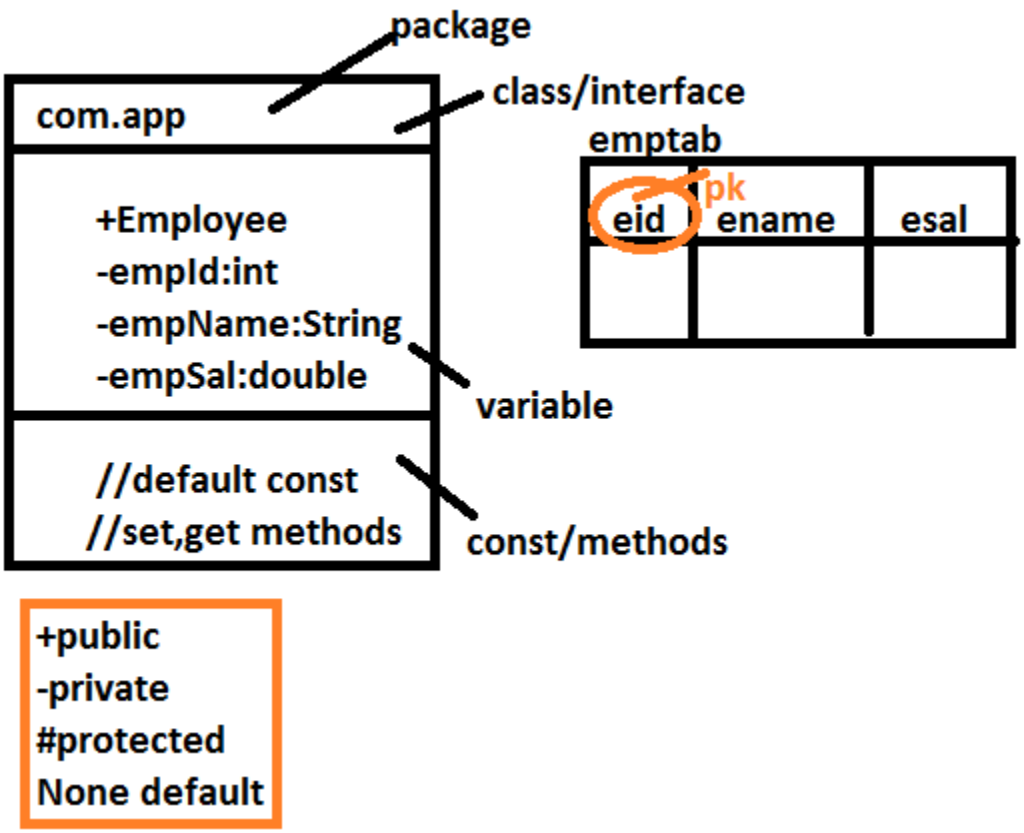
javabyraghu@gmail.c

javabyraghu@gmail.com



javabyraghu

Hibernate First App: -



out

javabyraghu@gmail.com

m ch gm g ch-

m ch gm

g ch ch m ch gm ch ch
g ch ch m ch gm m

g ch ch g ch

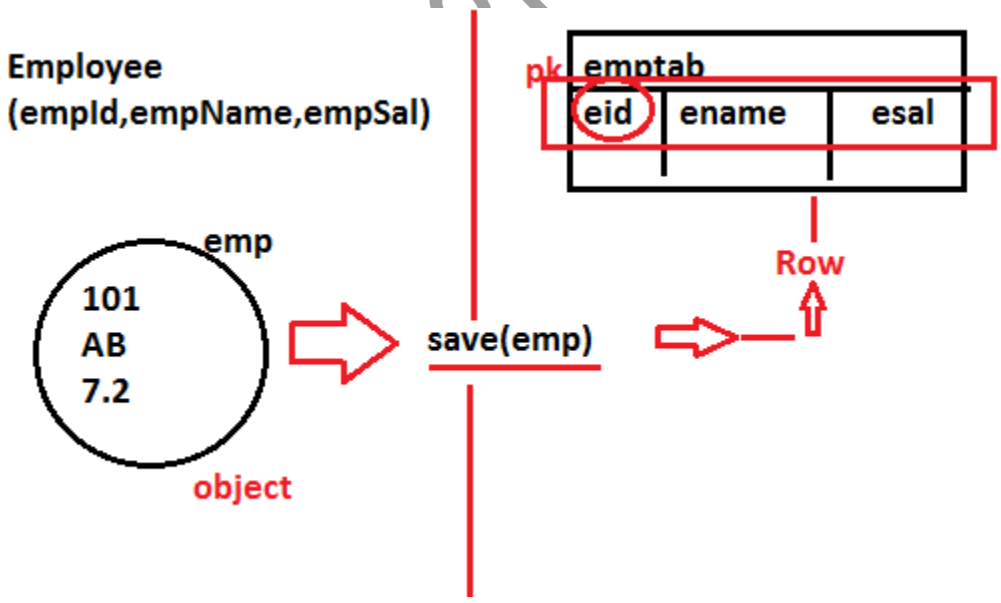
g ch ch m -
g ch ch m

m ch m ch m ch h

javabyraghu@gmail.com

javabyraghu@gmail.com

thu@gmail.com



javabyraghu@gmail.com

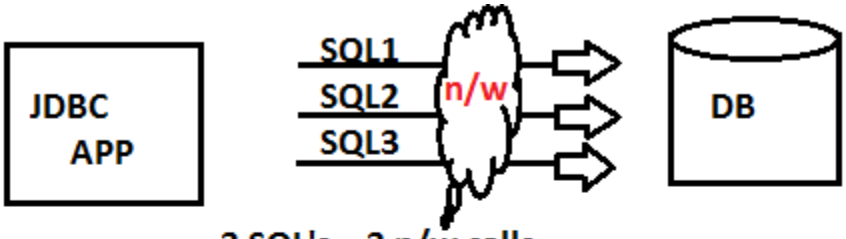
javabyraghu@gmail.com

javabyraghu@gmail.

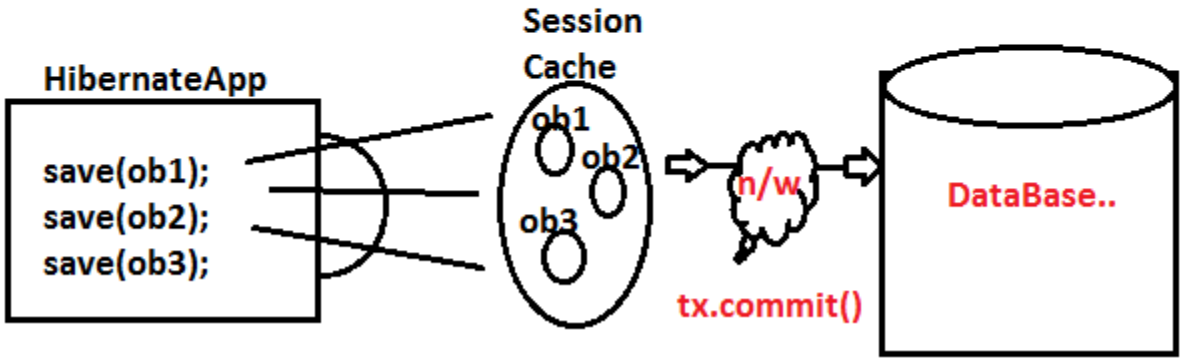
javabyraghu@gmail.com

javabyraghu@gmail.com


javabyraghu@gmail.c

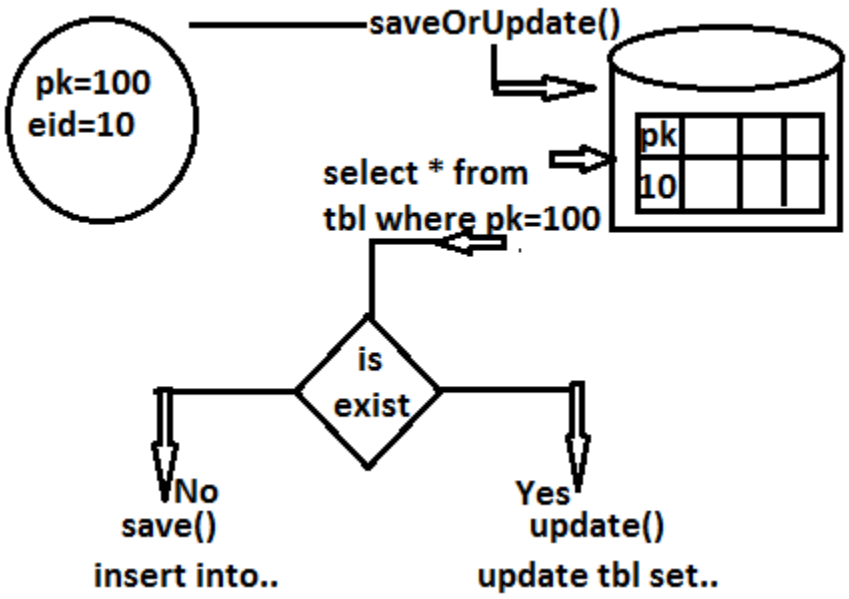


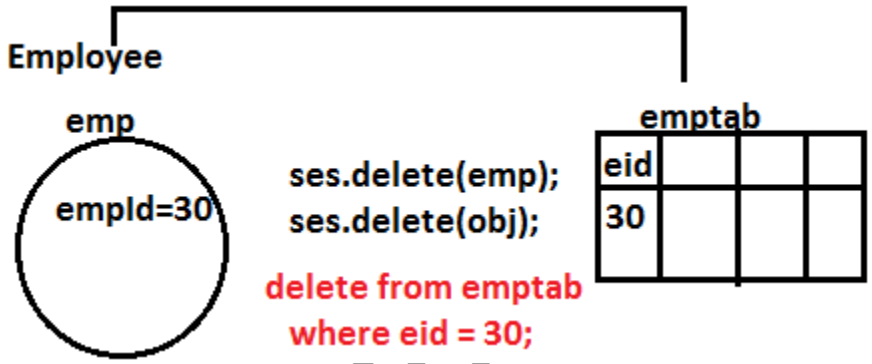
3 SQL's = 3 n/w calls



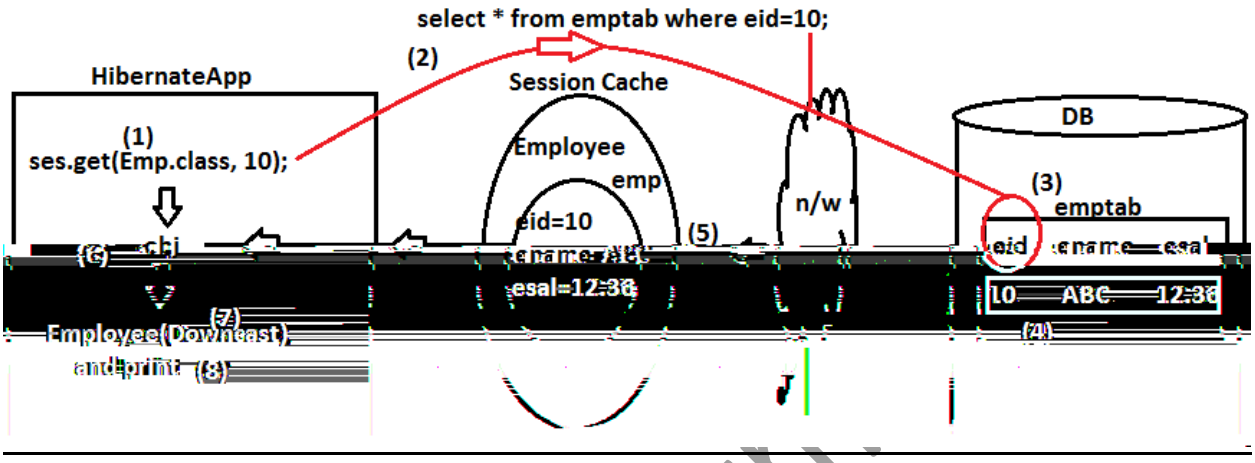
javabyraghu@gm

hu@gmail.com


saveOrUpdate()
pk=100
eid=10
pk
10
select * from
tbl where pk=100
is
exist
No
save()
insert into..
Yes
update()
update tbl set..



javabyraghu@gmail.com



out

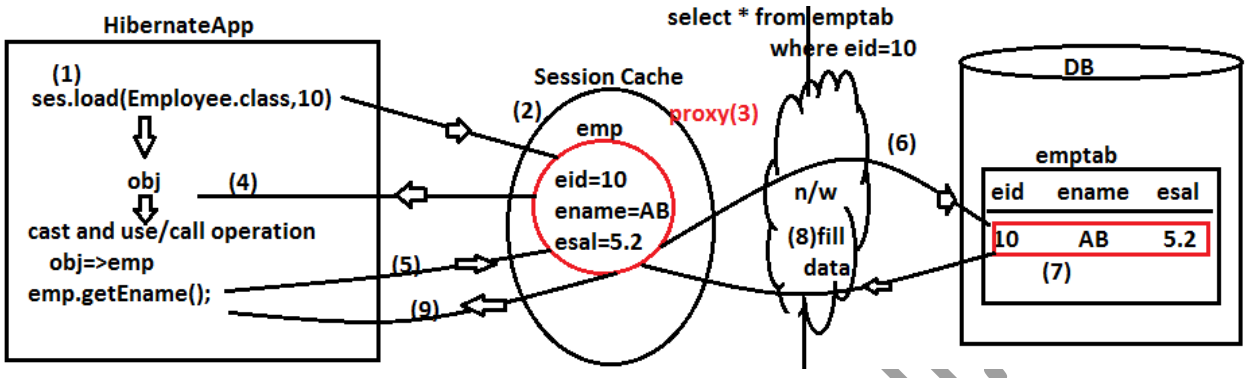
out

javabyraghu@gm

javabyraghu@gmail.com

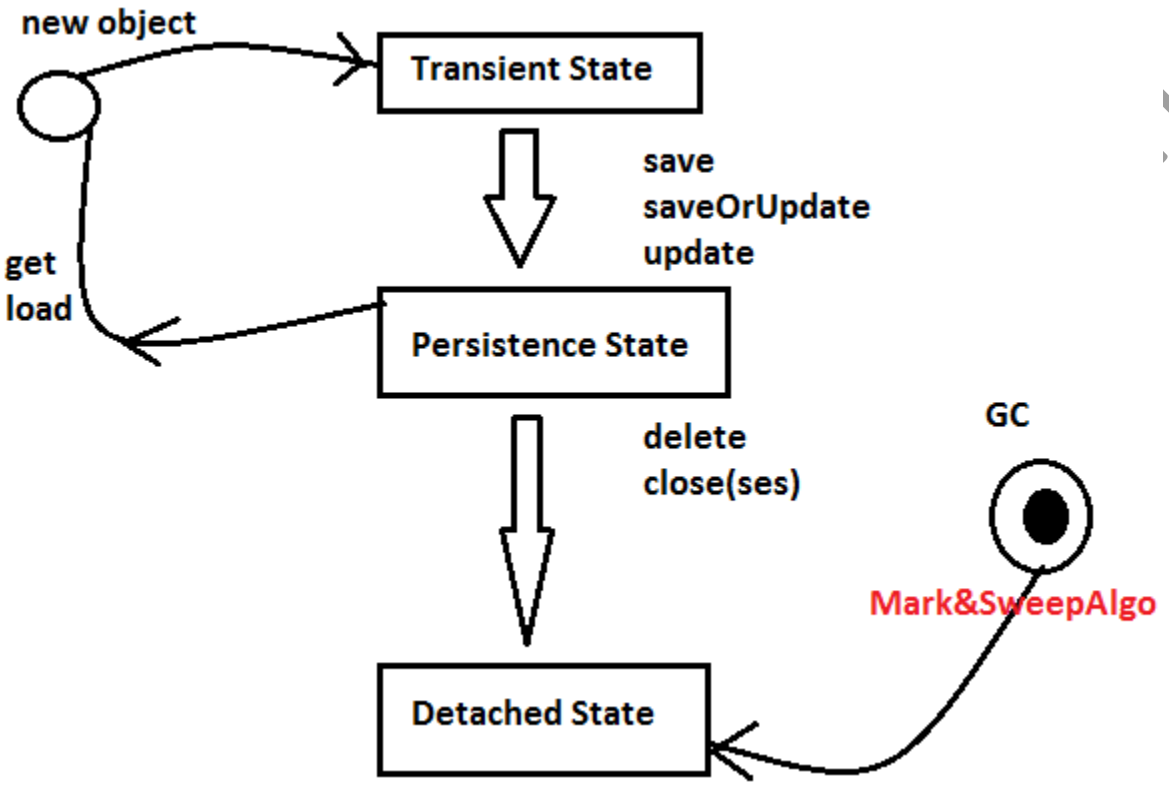
out

out



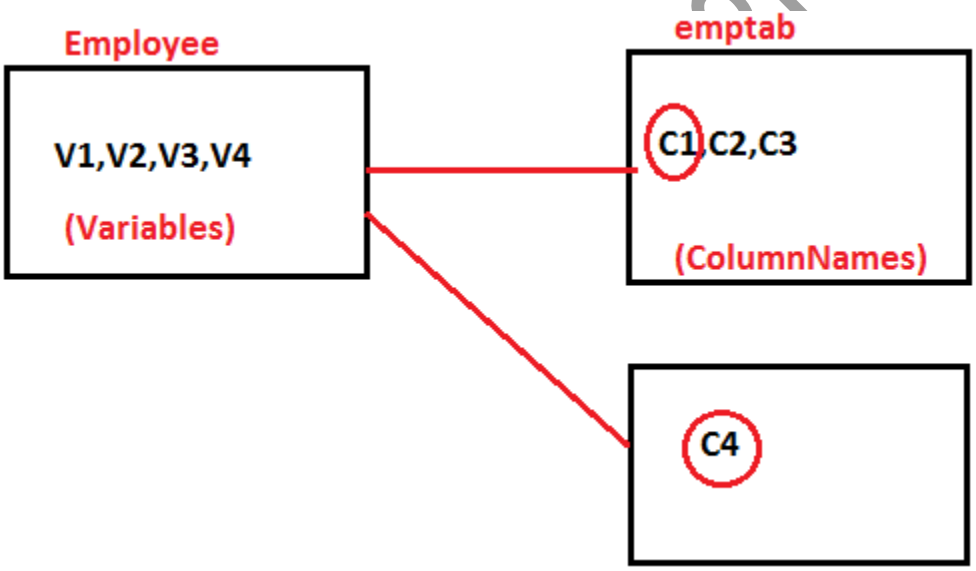
get()	load()
get hits DB always.	load hits cache always.
get returns null id record not exist.	load throws Exception if record not exist then (ObjectNotFoundException).
every time it makes network call.	only on reading data it makes network call.

javaby

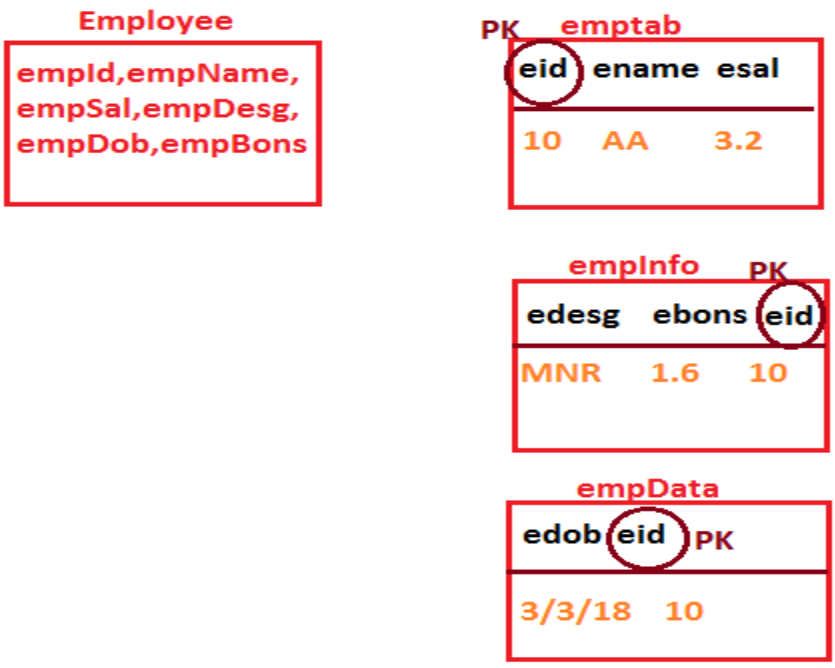

new object
Transient State
save
saveOrUpdate
update
get
load
Persistence State
GC
delete
close(ses)
Mark&SweepAlgo
Detached State

javabyraa

javabyraghu@gmail.com


Employee
V1,V2,V3,V4
(Variables)
emptab
C1,C2,C3
(ColumnNames)
C4

gmail.com


Employee
empId,empName,
empSal,empDesg,
empDob,empBons
PK
emptab
eid
ename
esal
10
AA
3.2
empInfo
PK
edesg
ebons
eid
MNR
1.6
10
empData
edob
eid
PK
3/3/18
10

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

→

→

→

→

javabyraghu@gmail.com

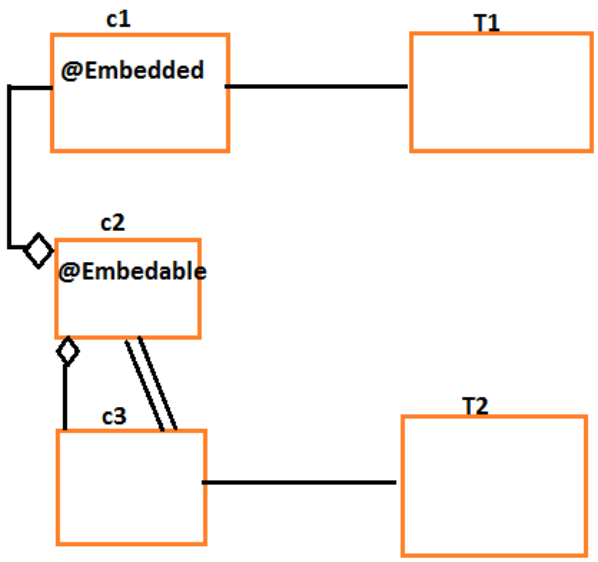
javabyraghu@gm

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com



Employee
empId,EmpName
empSal

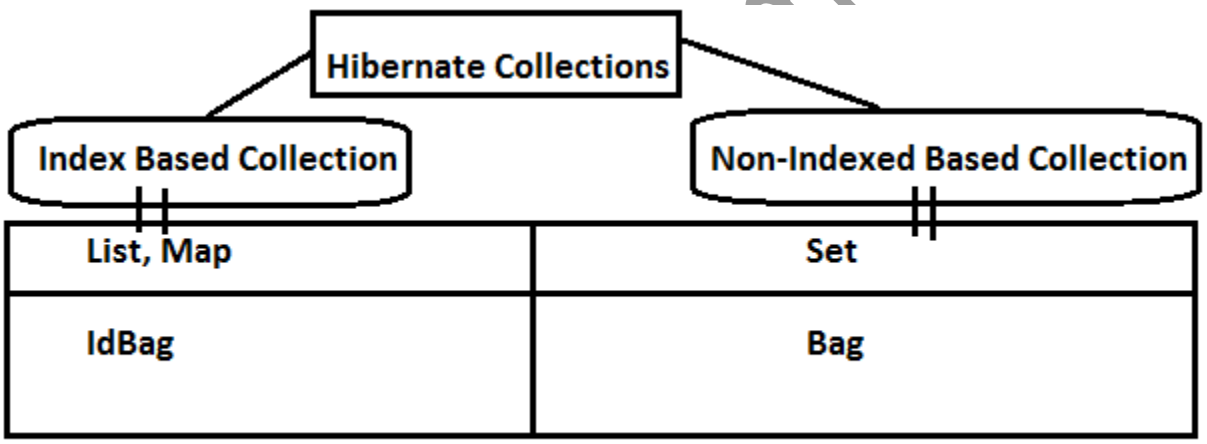
Address
addrId,loc

emptab
id,nm,sal,add,loc

javabyraghu@gma

javabyraghu@gmail.com

javabyraghu@gmail.com



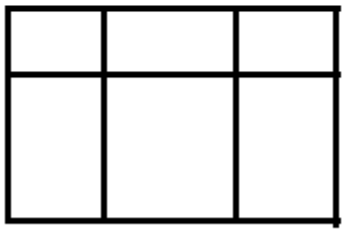
** Bag and IdBag are type "List" Internally design is given from Hibernate F/w

javaby mail.com

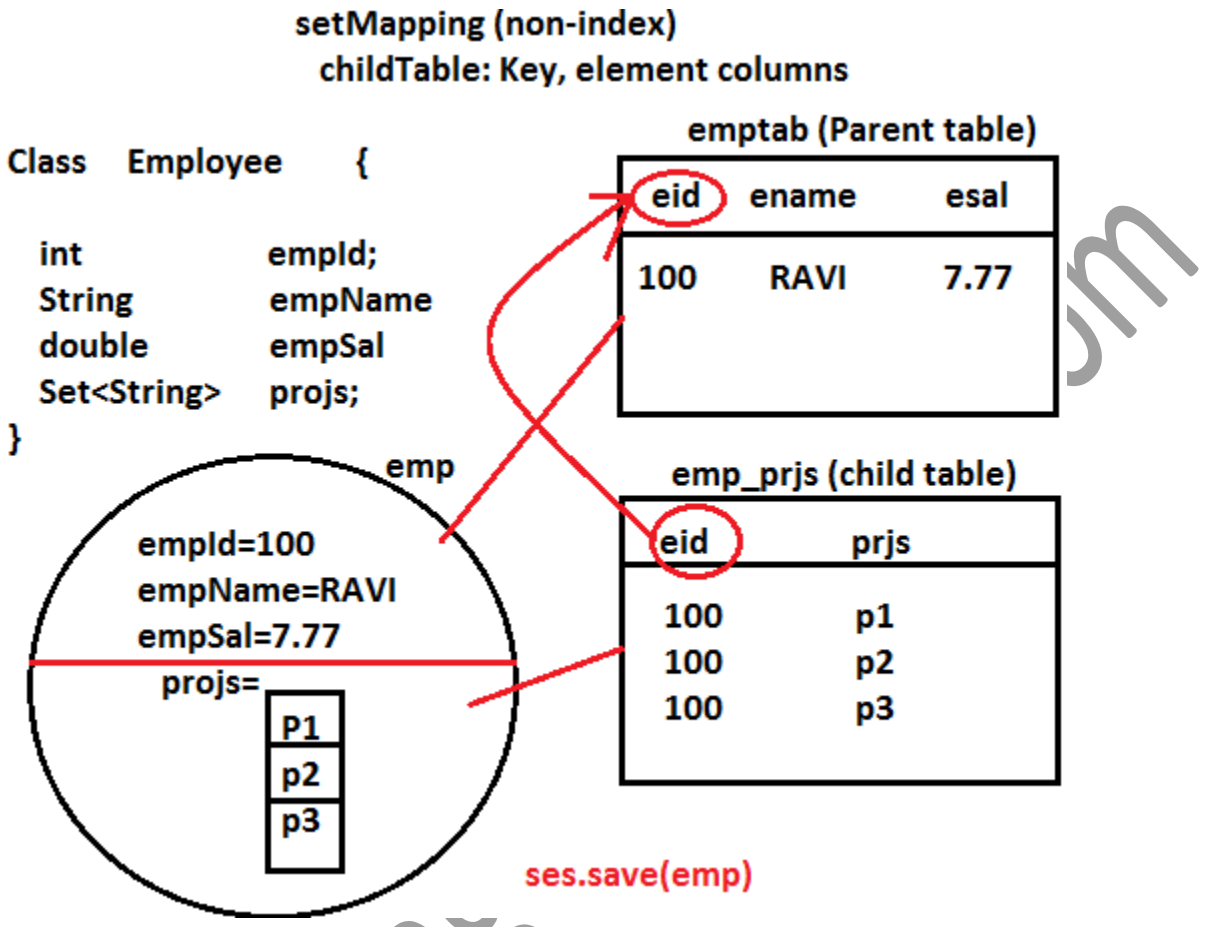
javabyraghu@gmail.com

hu@gmail.com

Hibernate	DB
Class	Table
Primitive Variable	one column
collectionType Variable	childTable

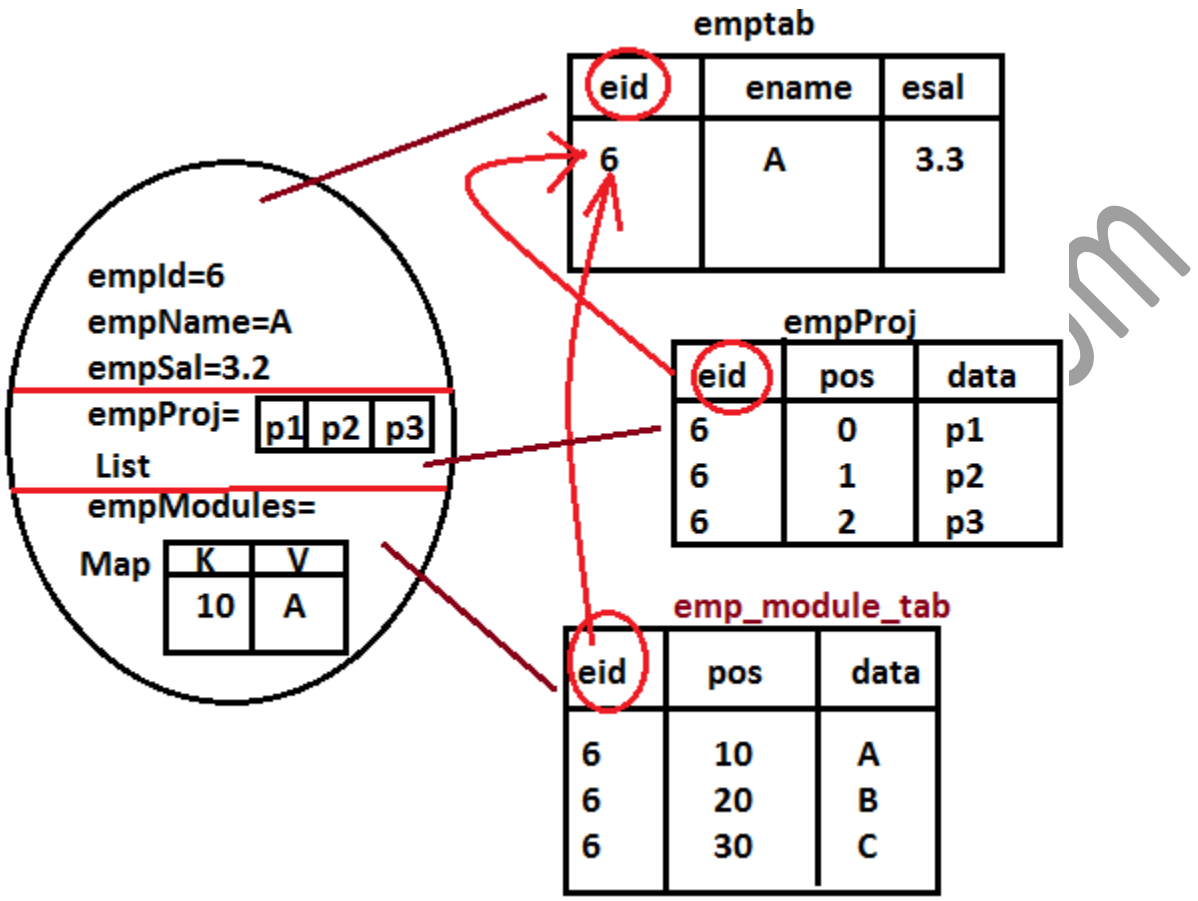


Key column index column element column

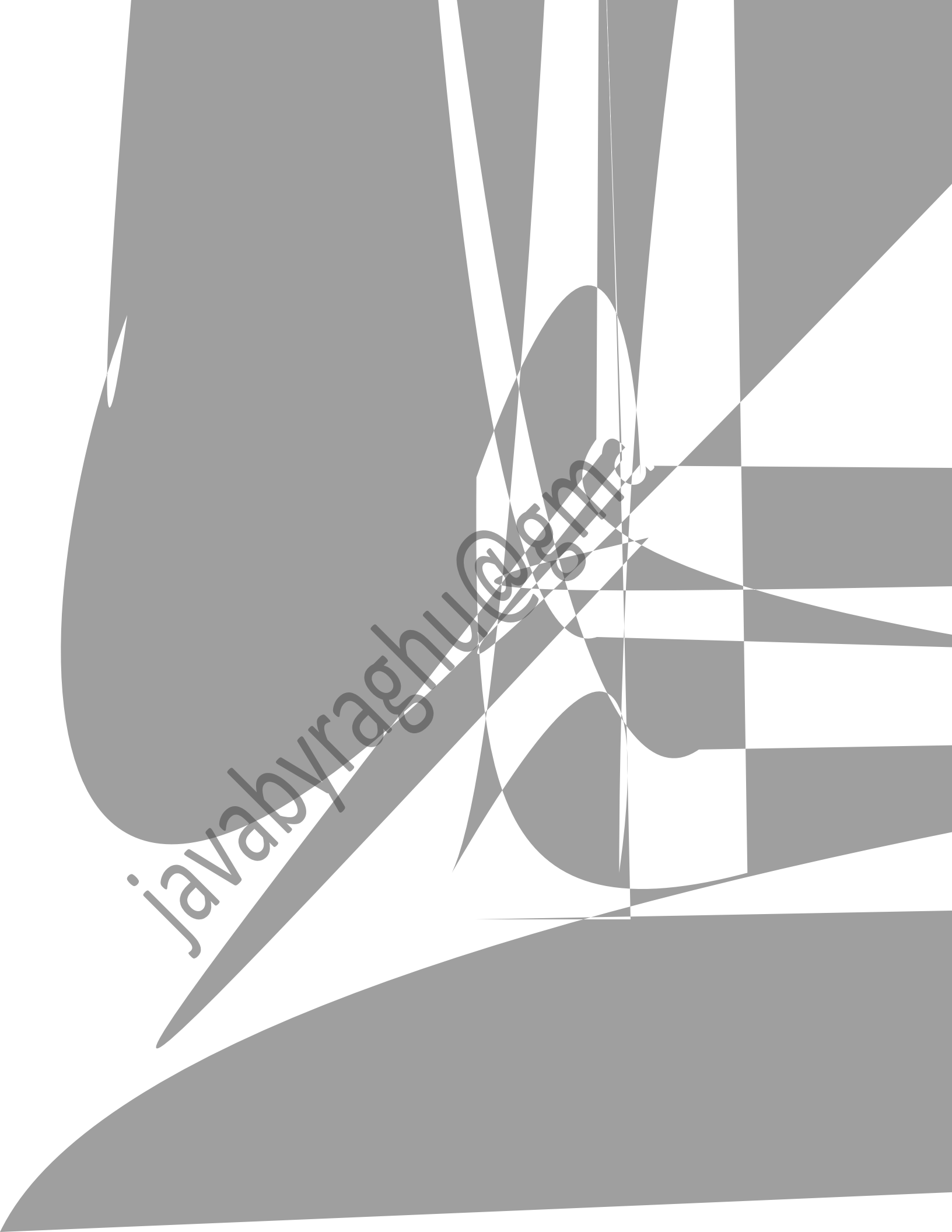


javabyraj.com

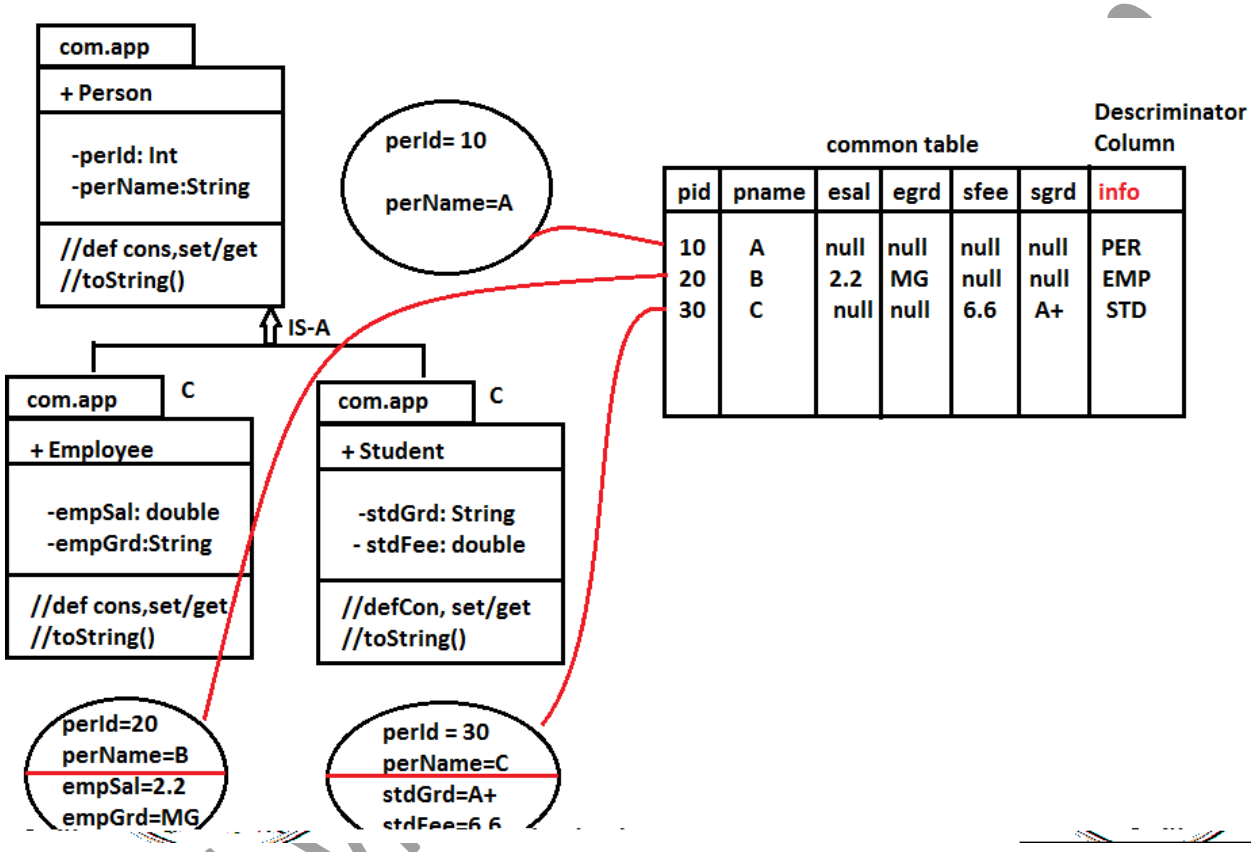
javabyraghu@gmail.com


emptab
eid
ename
esal
6
A
3.3
empId=6
empName=A
empSal=3.2
empProj=
p1
p2
p3
List
empModules=
Map
K
V
10
A
empProj
eid
pos
data
6
0
p1
6
1
p2
6
2
p3
emp_module_tab
eid
pos
data
6
10
A
6
20
B
6
30
C

javabyraghu@gmail.


javabyraghu@gmail

javabyraghu@gmail.c


com.app
+ Person
-perId: Int
-perName:String
//def cons,set/get
//toString()
IS-A
com.app
C
+ Employee
-empSal: double
-empGrd:String
//def cons,set/get
//toString()
com.app
C
+ Student
-stdGrd: String
- stdFee: double
//defCon, set/get
//toString()
perId= 10
perName=A
perId=20
perName=B
empSal=2.2
empGrd=MG
perId = 30
perName=C
stdGrd=A+
common table
Descriminator Column
pid pname esal egrd sfee sgrd info
10 A null null null null PER
20 B 2.2 MG null null EMP
30 C null null 6.6 A+ STD

javaby

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

I

javabyraghu@gmail.com

javabyraghu@gmail.com

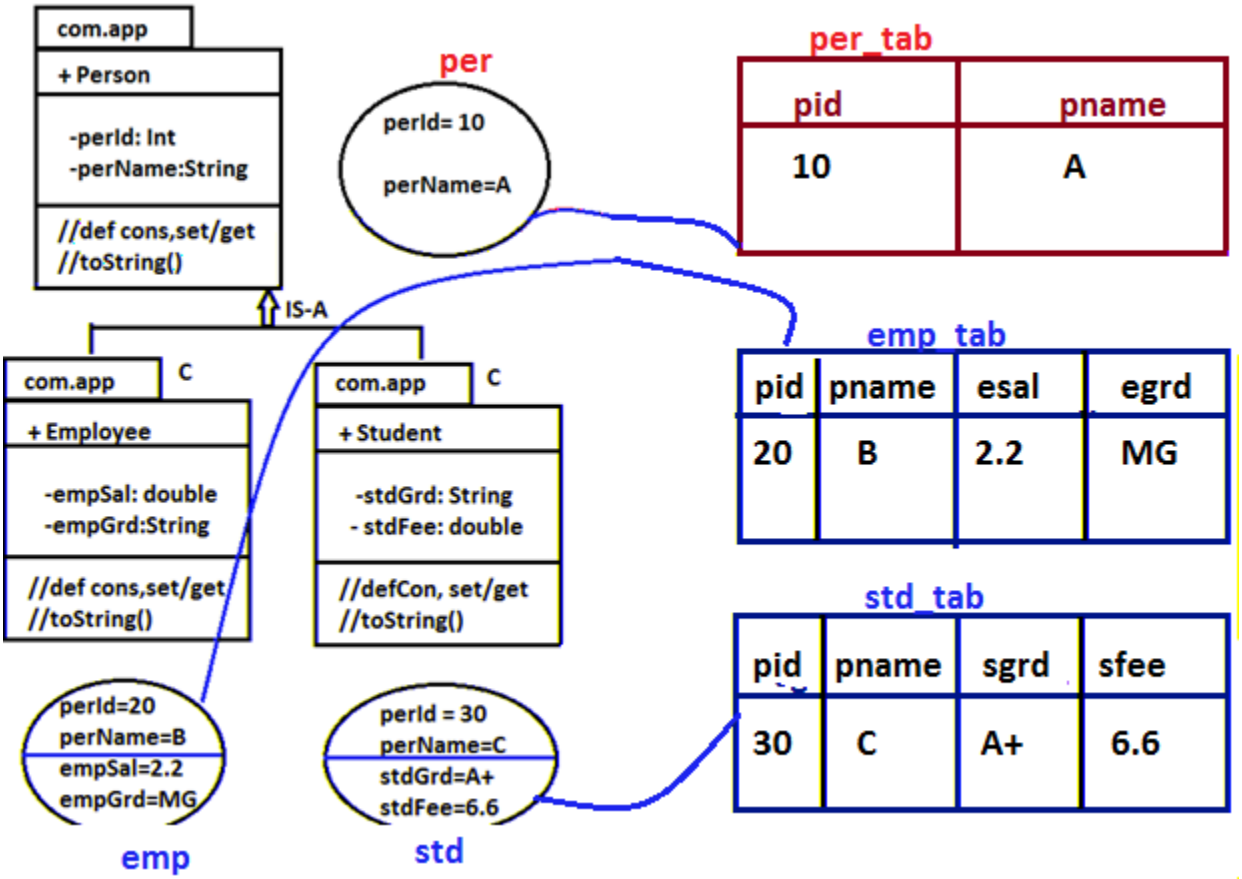
javabyraghu@gmail.com

javabyraghu@gm

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gma


com.app
+ Person
-perId: Int
-perName:String
//def cons,set/get
//toString()
IS-A
com.app
C
+ Employee
-empSal: double
-empGrd:String
//def cons,set/get
//toString()
com.app
C
+ Student
-stdGrd: String
- stdFee: double
//defCon, set/get
//toString()
per
perId= 10
perName=A
perId=20
perName=B
empSal=2.2
empGrd=MG
emp
perId = 30
perName=C
stdGrd=A+
stdFee=6.6
std
per_tab
pid
pname
10
A
emp_tab
pid
pname
esal
egrd
20
B
2.2
MG
std_tab
pid
pname
sgrd
sfee
30
C
A+
6.6

javabyt

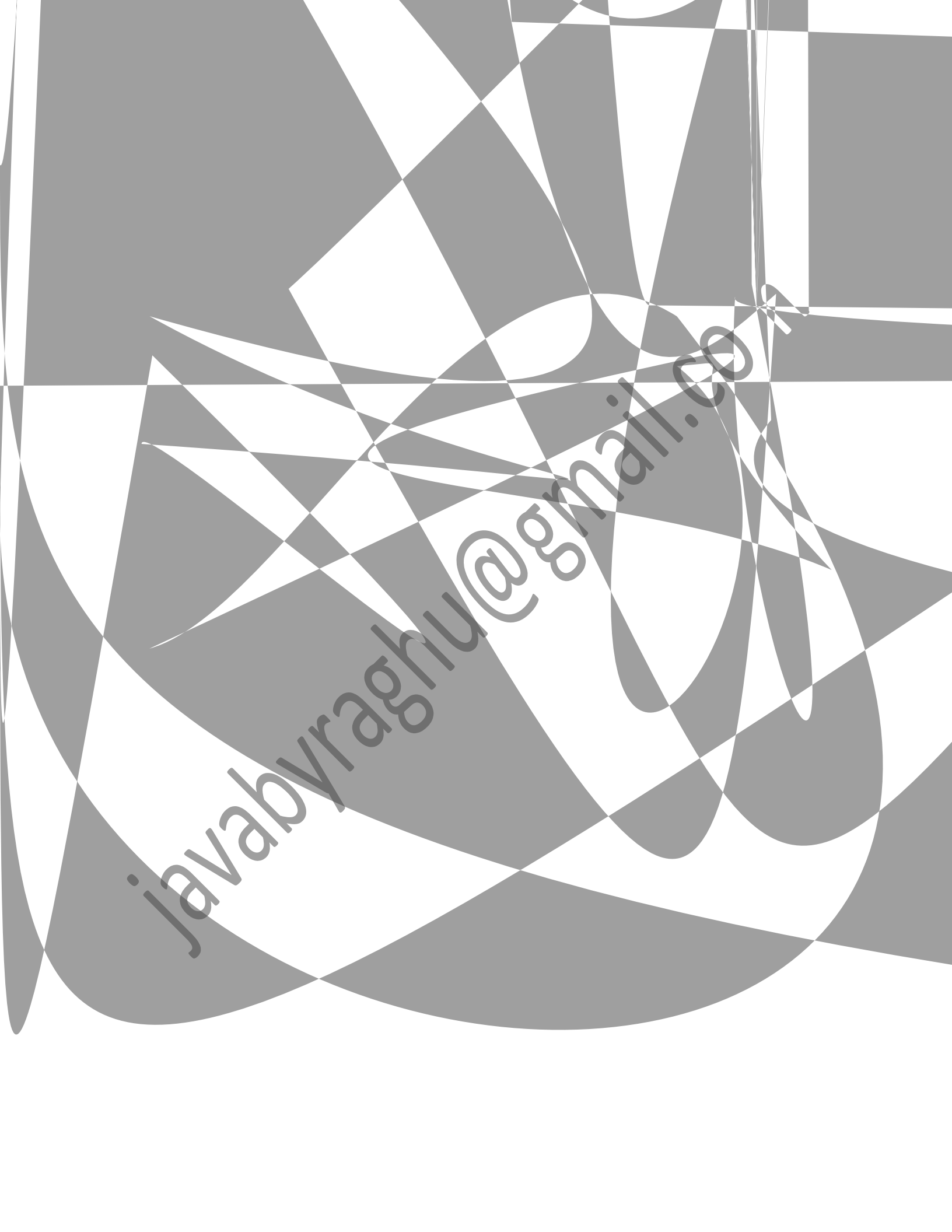
javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

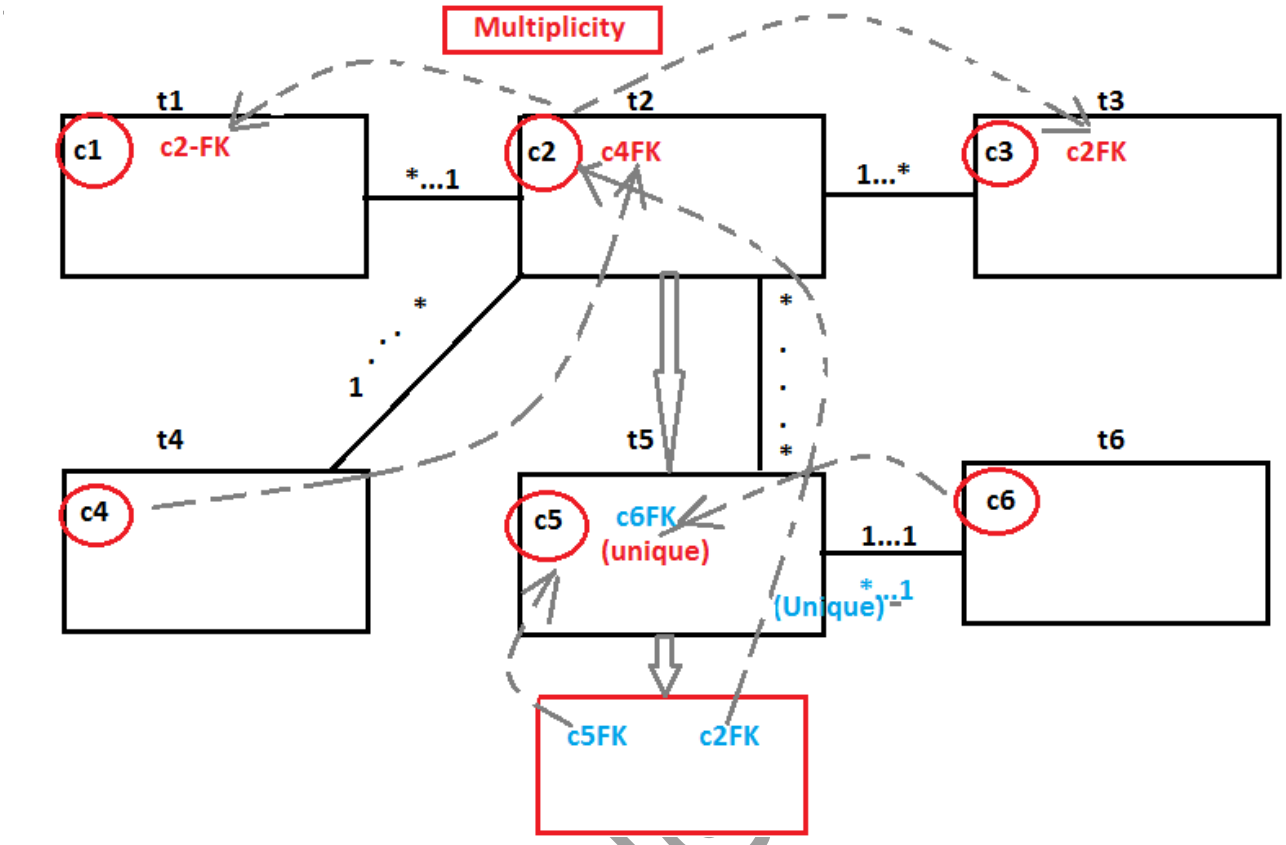
javabyraghu@gm


javabyraghu@gmail.com

I

javabyraghu@gmail.com

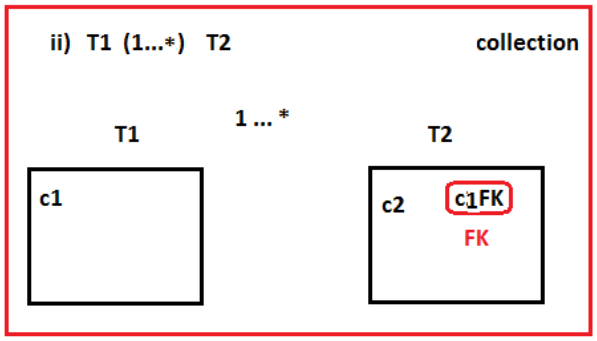
javabyraghu@gmail.com

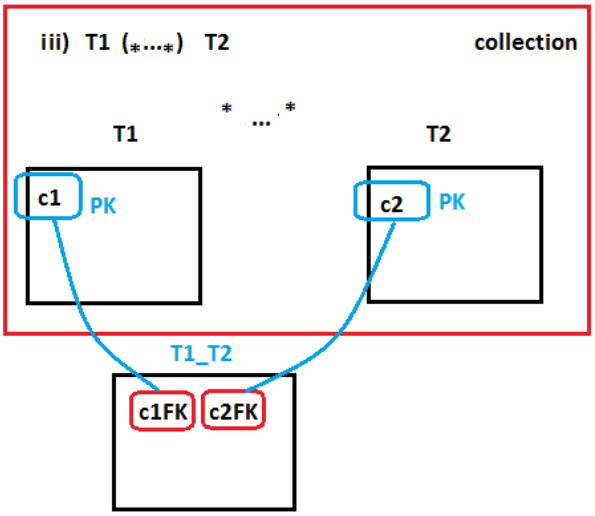


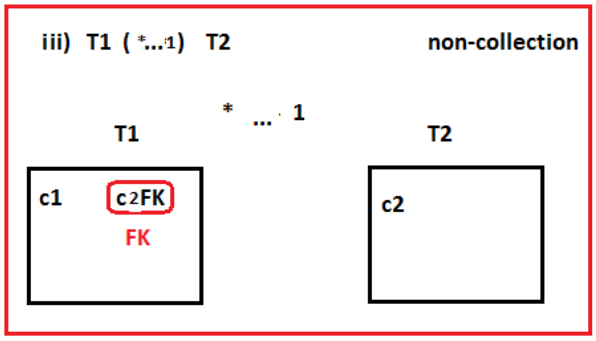
-

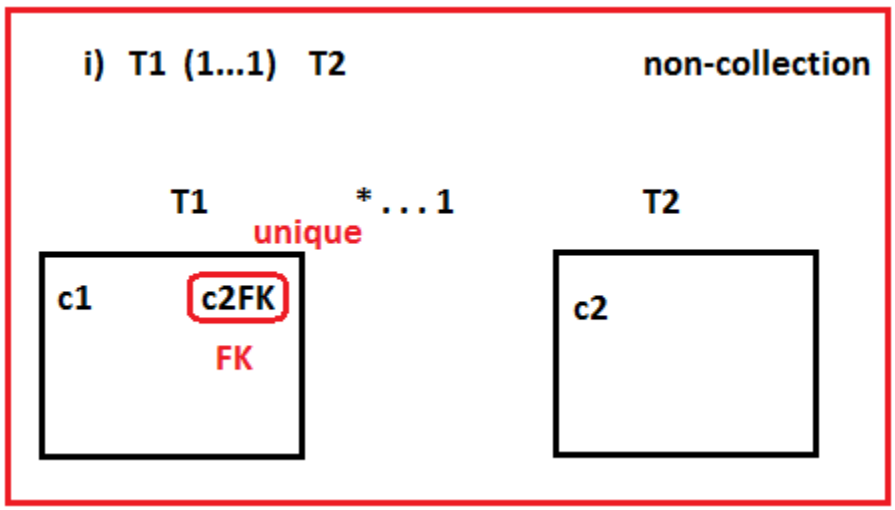
Non-Collection	Collection
1...1	1...*
*...1	*...*

-
- |


ii) T1 (1...*) T2
collection
T1
1 ... *
T2
c1
c2
c1FK
FK


iii) T1 (*...*) T2
collection
* ... *
T1
T2
c1
PK
c2
PK
T1_T2
c1FK
c2FK


iii) T1 (*...1) T2
non-collection
* ... 1
T1
T2
c1
c2FK
FK
c2

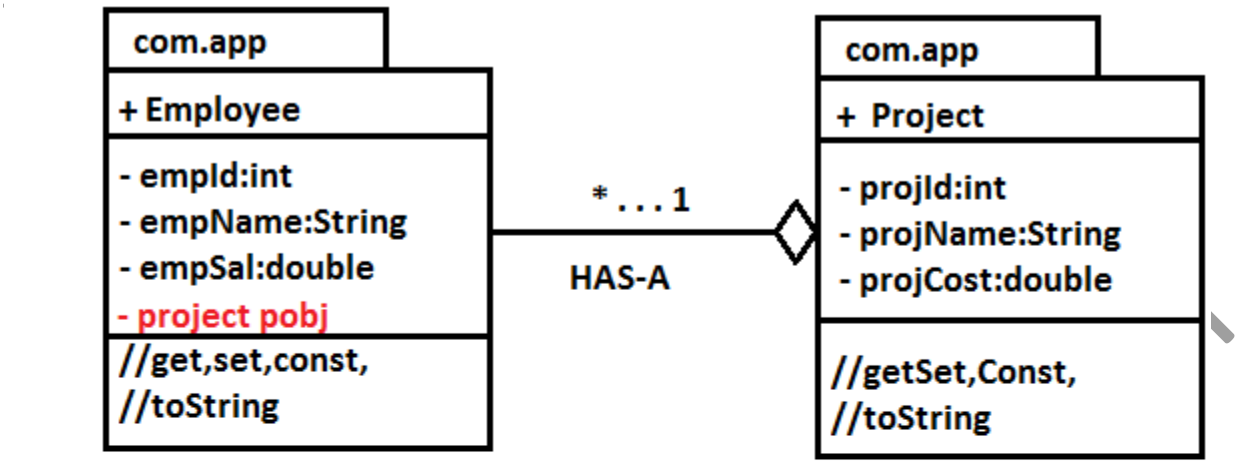

i) T1 (1...1) T2
non-collection
T1
* . . . 1
T2
unique
c1
c2FK
FK
c2

- →

javabyraghu@gmail.com

→

javabyraghu@gmail.com



pk FK * . . . 1 pk

eid	ename	esal	pidFK
10	AAA	2.2	888
11	BBB	3.3	888
12	CCC	4.4	888

extra column

pid	pname	pcost
888	p1	99.99

javabyraghu

javabyraghu@gmail.com

I

javabyraghu@gmail.com

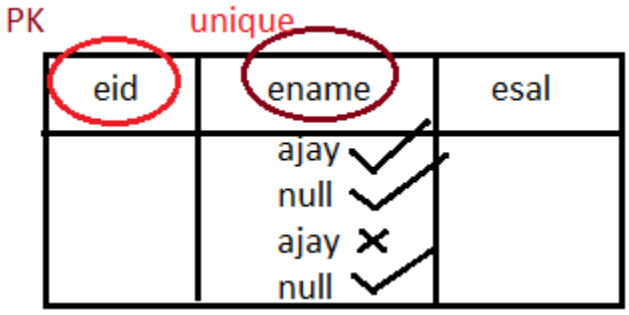
javabyraghu@gmail.com

javabyraghu@gmail.com

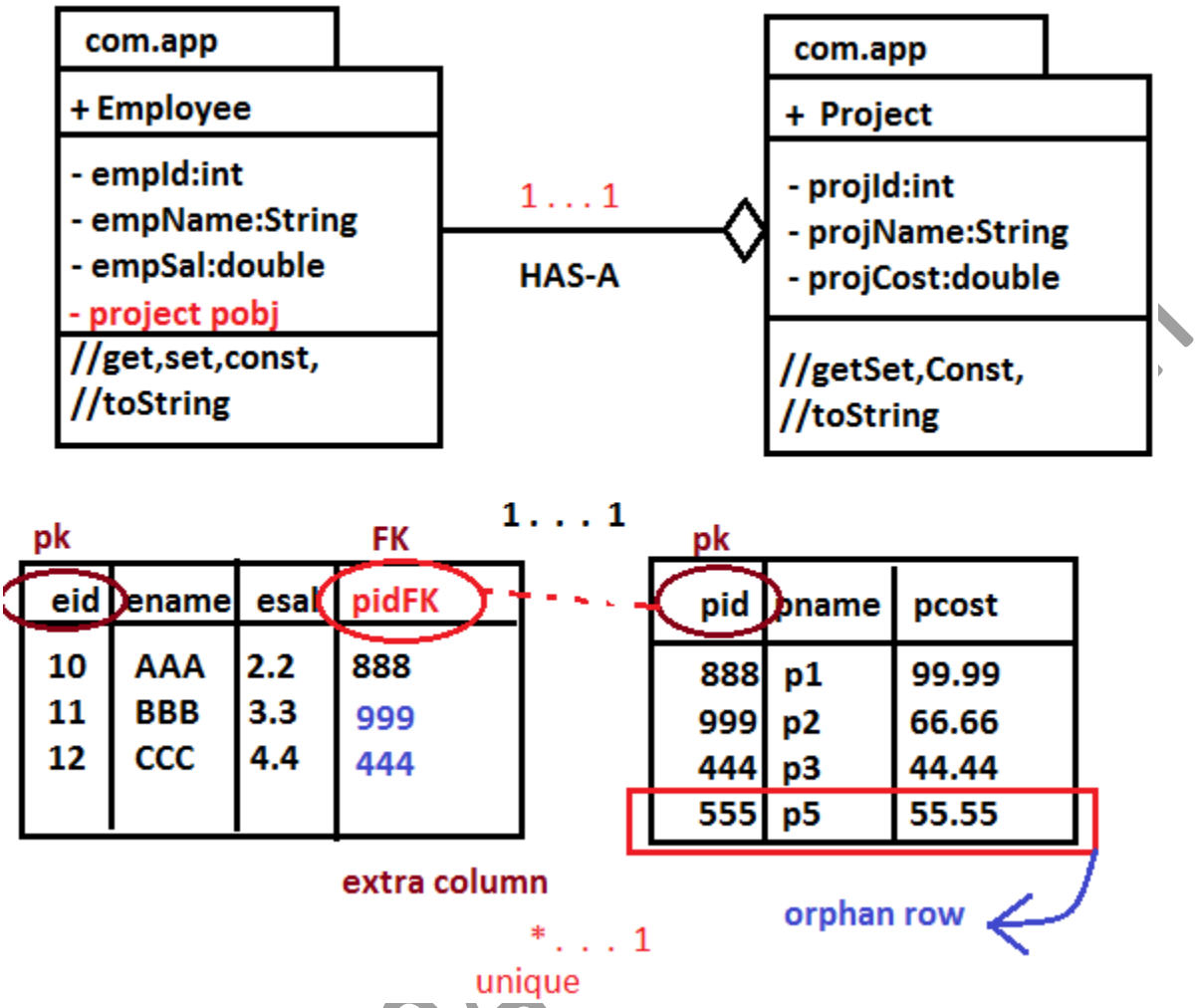
javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com



javabyraghu@gmail.com


com.app
+ Employee
- empId:int
- empName:String
- empSal:double
- project pobj
//get,set,const,
//toString
1 . . . 1
HAS-A
com.app
+ Project
- projId:int
- projName:String
- projCost:double
//getSet,Const,
//toString
1 . . . 1
pk
FK
eid
ename
esal
pidFK
10
AAA
2.2
888
11
BBB
3.3
999
12
CCC
4.4
444
pk
pid
pname
pcost
888
p1
99.99
999
p2
66.66
444
p3
44.44
555
p5
55.55
extra column
orphan row
* . . . 1
unique

javabyraa

I

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

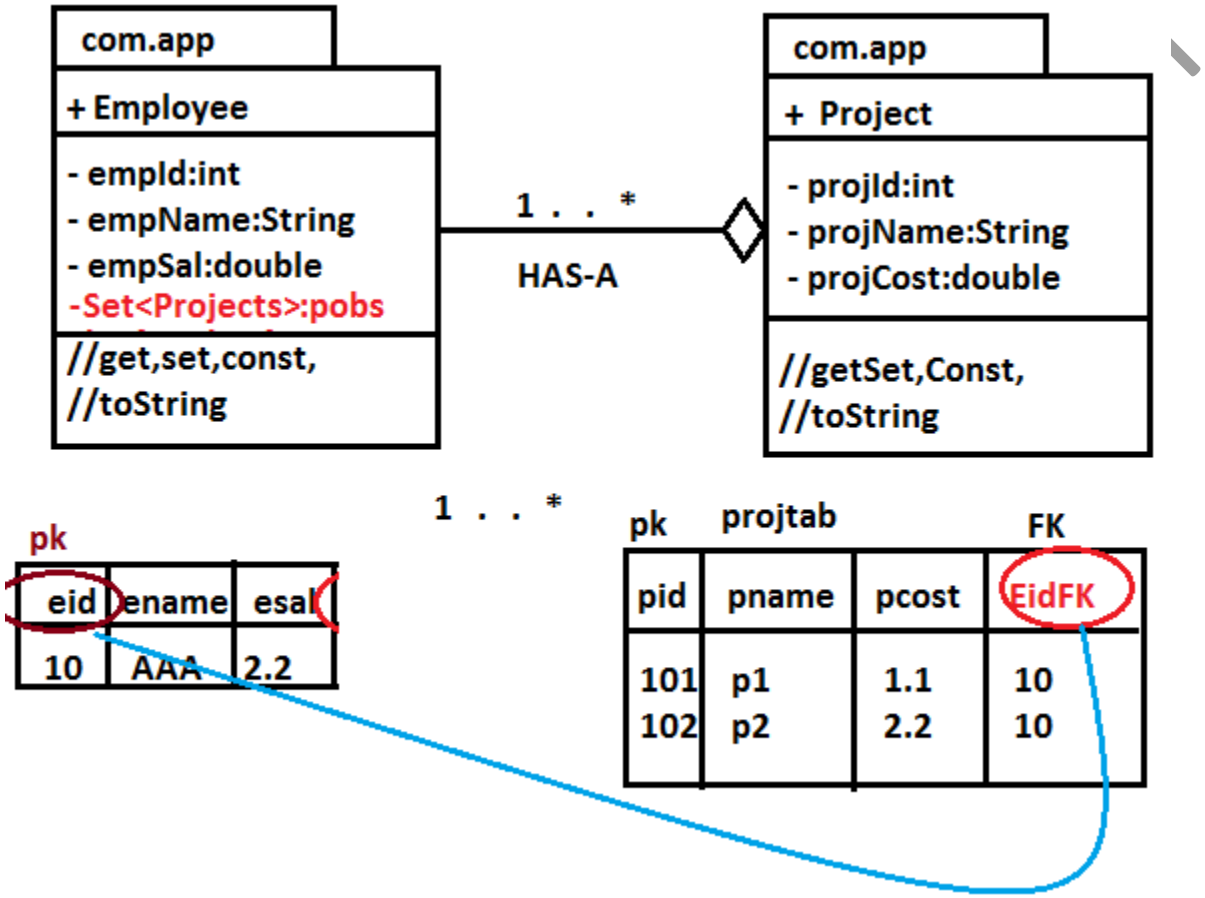
javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com


com.app
+ Employee
- empId:int
- empName:String
- empSal:double
-Set<Projects>:pobs
//get,set,const,
//toString
1 . . *
HAS-A
com.app
+ Project
- projId:int
- projName:String
- projCost:double
//getSet,Const,
//toString
1 . . *
pk
eid
ename
esal
10
AAA
2.2
pk
projtab
FK
pid
pname
pcost
EidFK
101
p1
1.1
10
102
p2
2.2
10


javabyraghu@gmail

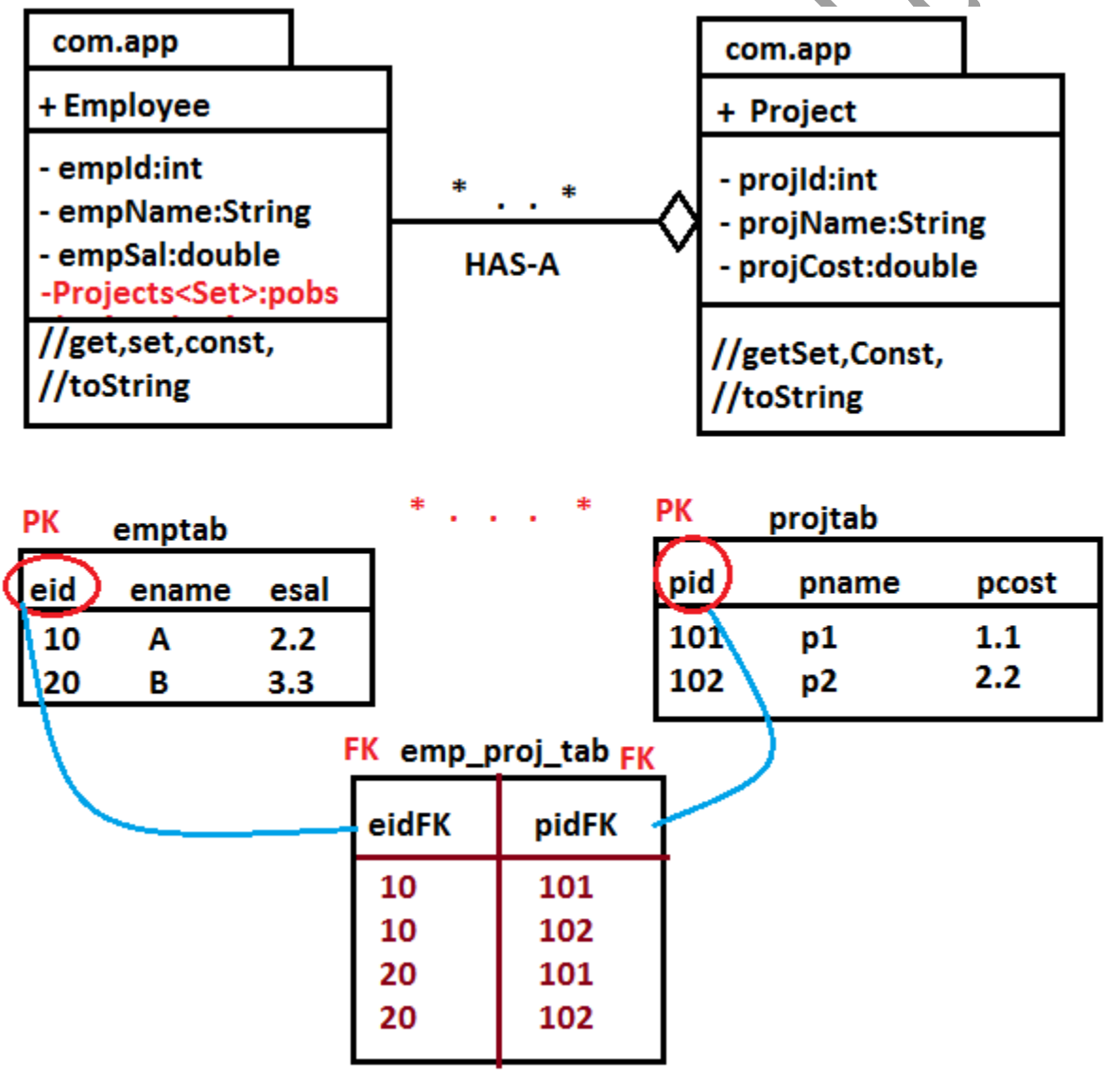
javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gm

javabyraghu@gmail.com

javabyraghu@gmail.com


com.app
+ Employee
- empId:int
- empName:String
- empSal:double
-Projects<Set>:pobs
//get,set,const,
//toString
* . . *
HAS-A
com.app
+ Project
- projId:int
- projName:String
- projCost:double
//getSet,Const,
//toString
PK
emptab
eid ename esal
10 A 2.2
20 B 3.3
* . . . *
PK
projtab
pid pname pcost
101 p1 1.1
102 p2 2.2
FK
emp_proj_tab
FK
eidFK pidFK
10 101
10 102
20 101
20 102

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

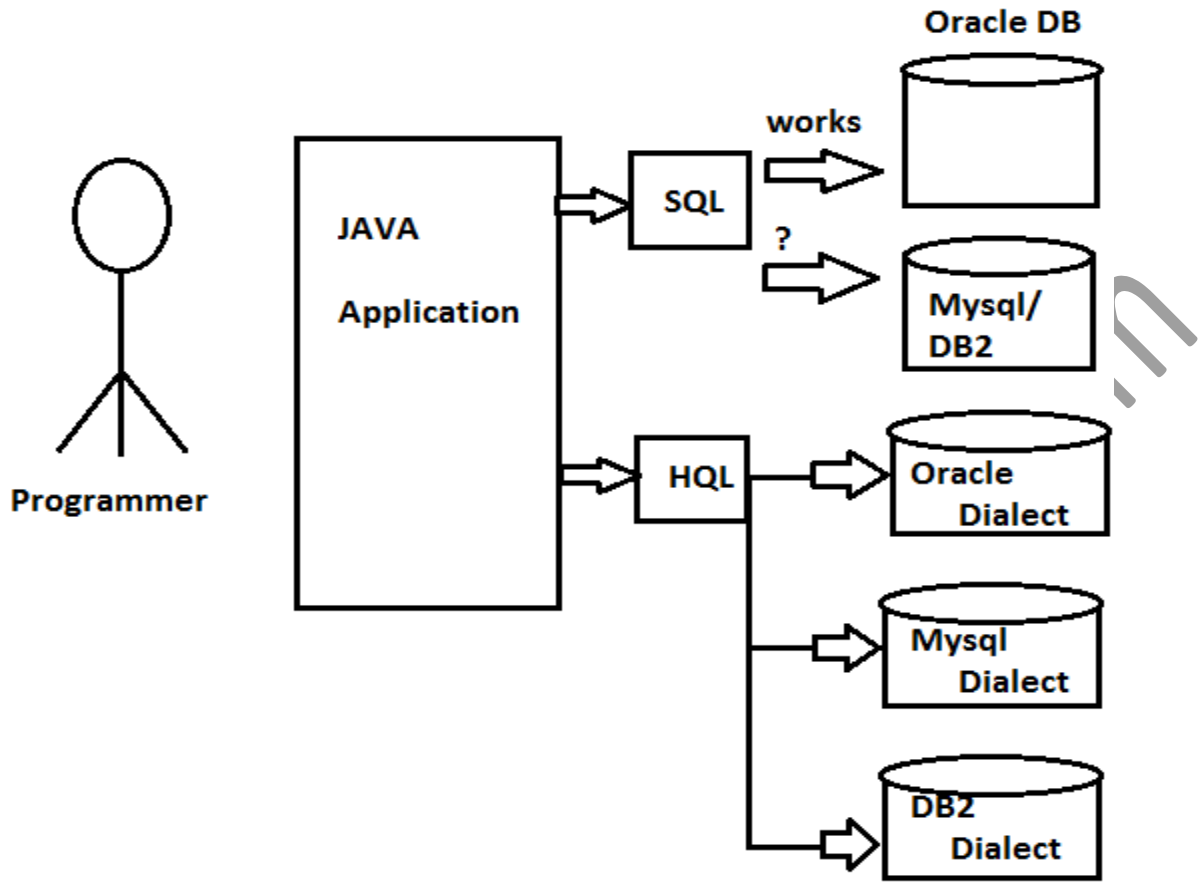
javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com


javabyraghu@gmail.com

javabyraghu@gmail.com



HQL Operations

select

insert

update

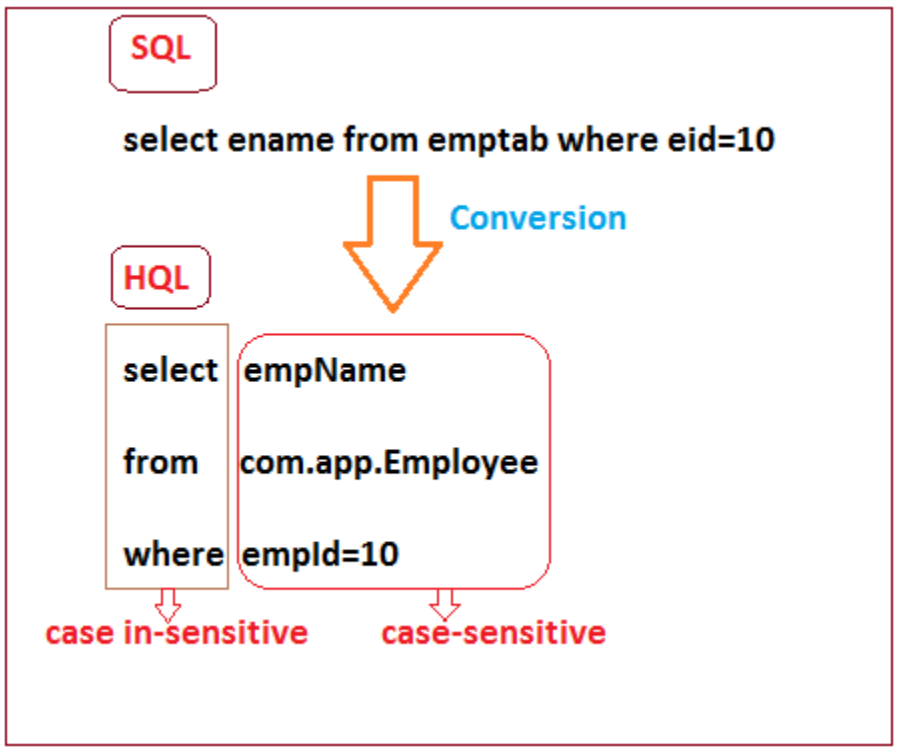
delete

copy

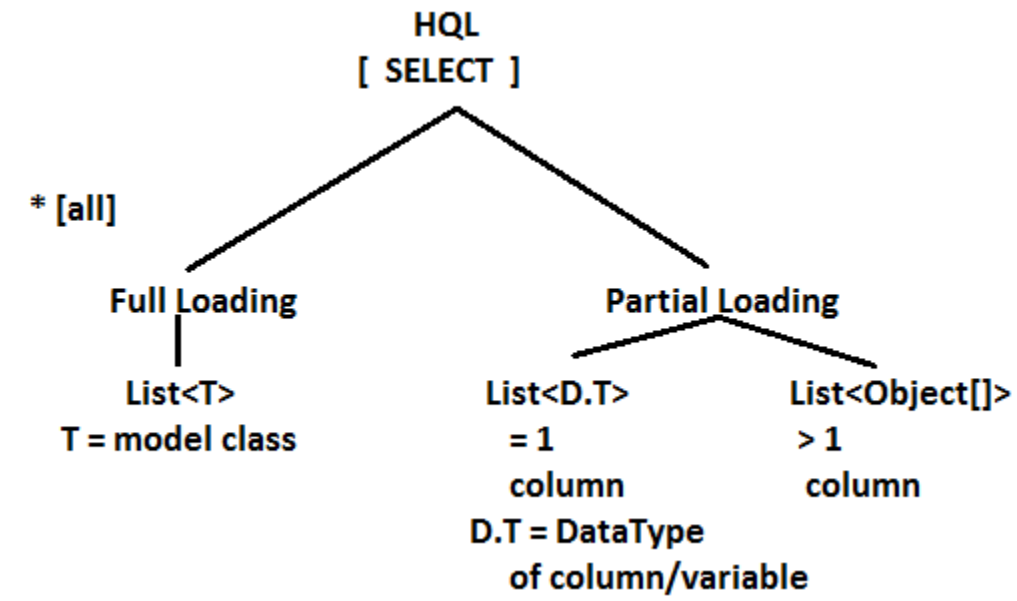
→

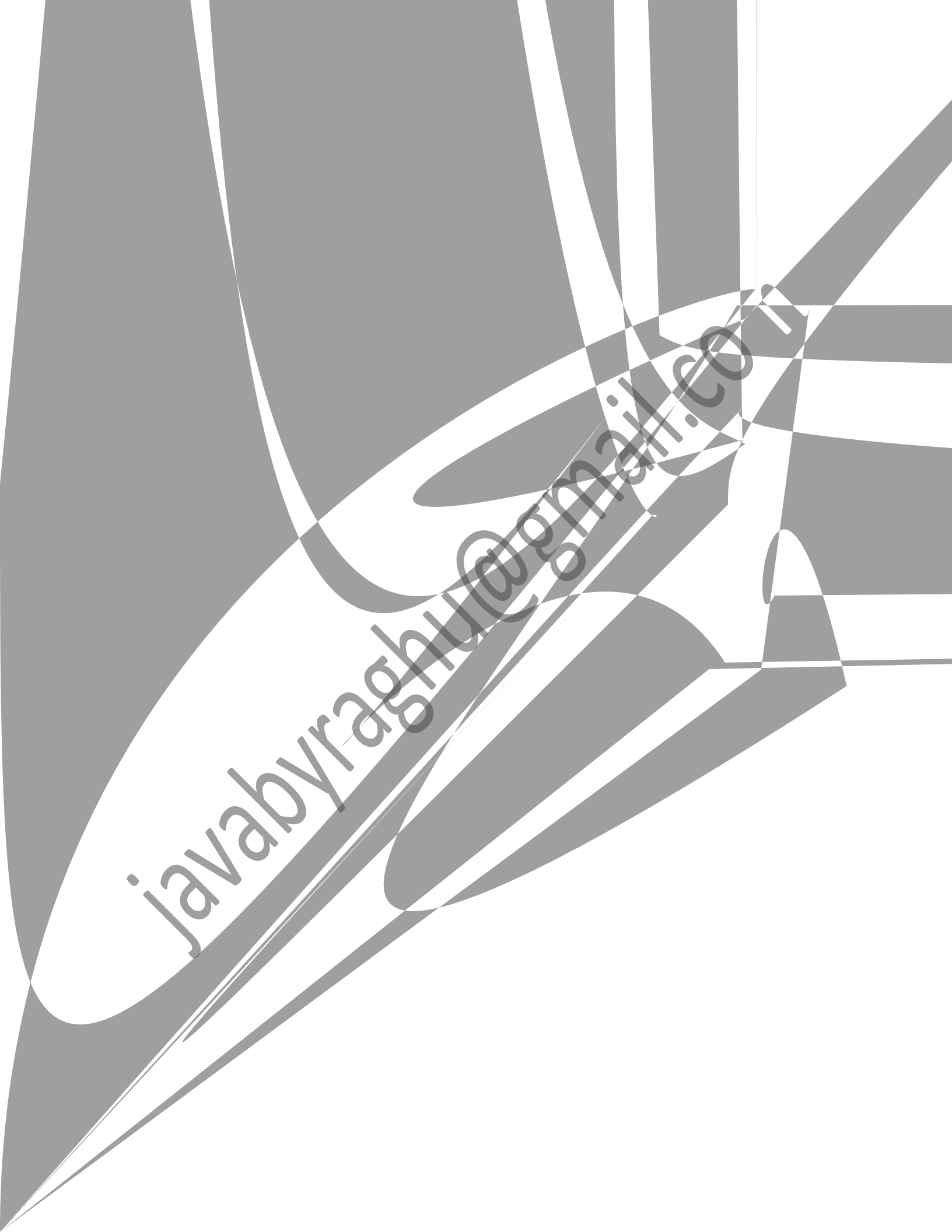
→

hu@gmail.com


SQL
select ename from emptab where eid=10
Conversion
HQL
select empName
from com.app.Employee
where empId=10
case in-sensitive
case-sensitive

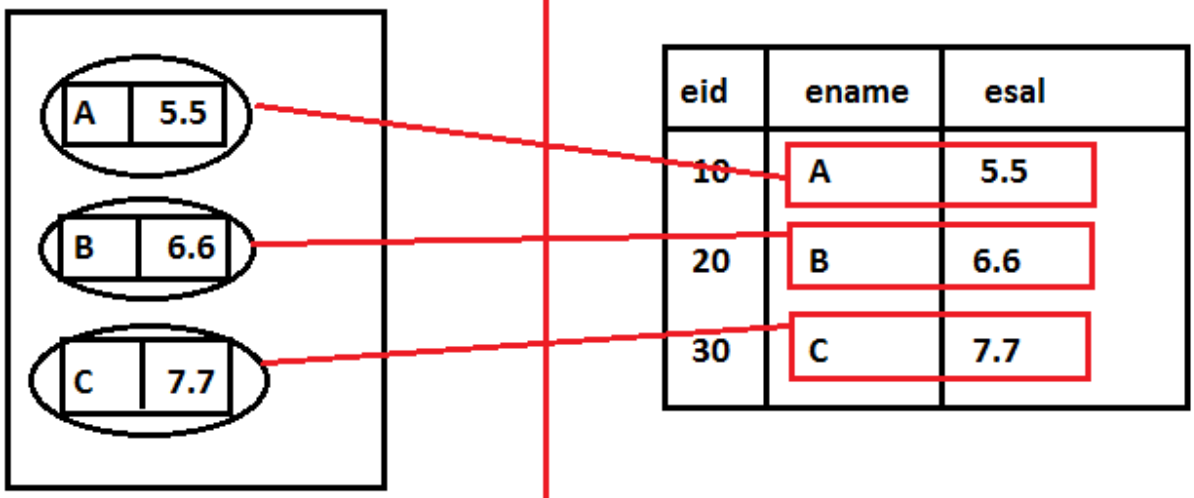
raghu@gmail.com




javabyraghu@gmail.com

3) Partial Loading with > 1 column

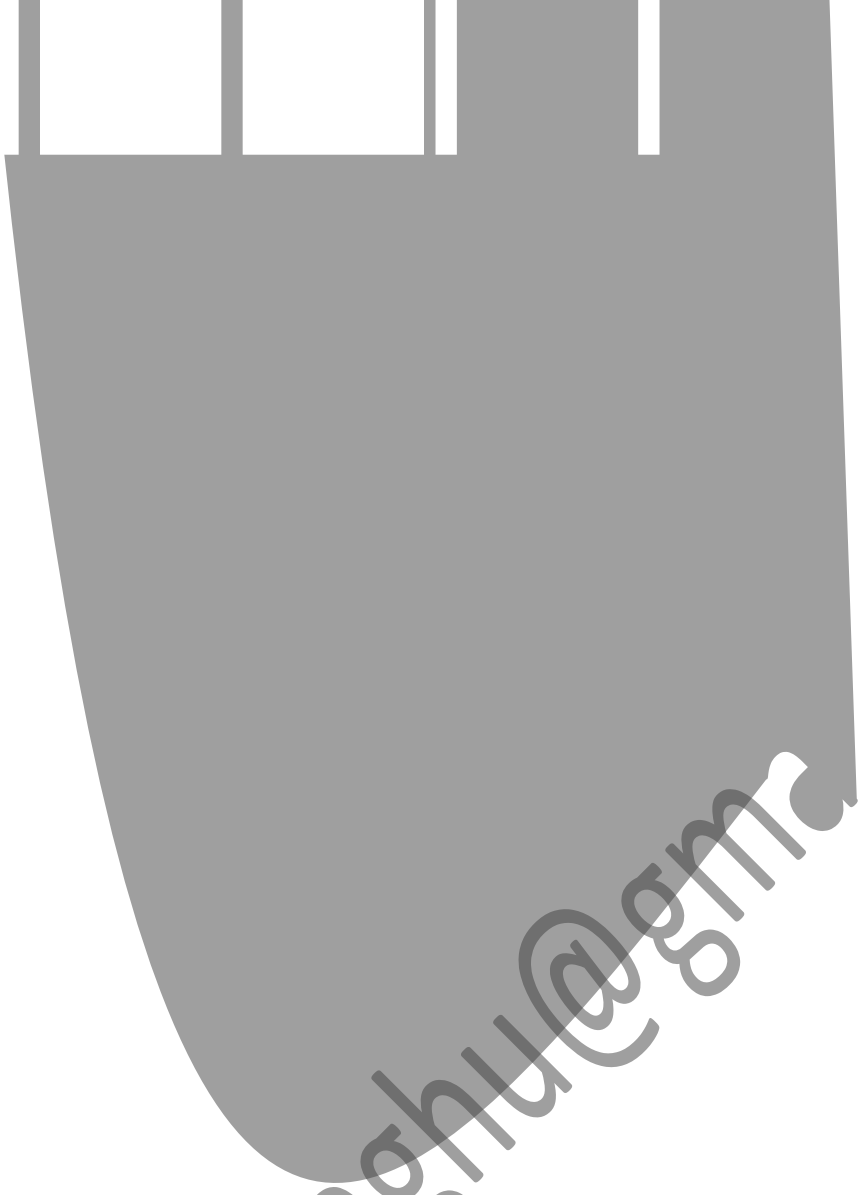
hql: select empName,empSal,
from com.app.Employee



javabyraghu

javabyraghu@gmail.com

javabyraghu@gmail.com



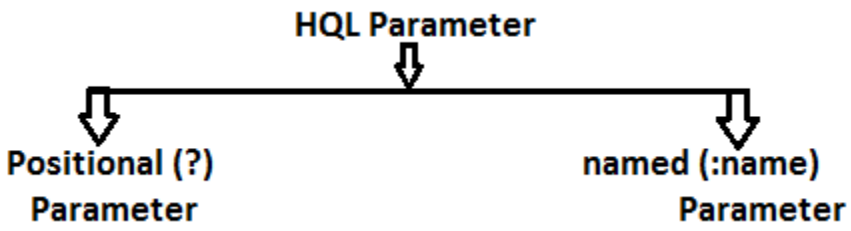
javabyraghu@gm

javabyraghu@gmail.com

I

javabyraghu@gmail.com

javabyraghu@gmail.com



Jagadeeraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.

javabyraghu@gmail.com

→

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

→

I

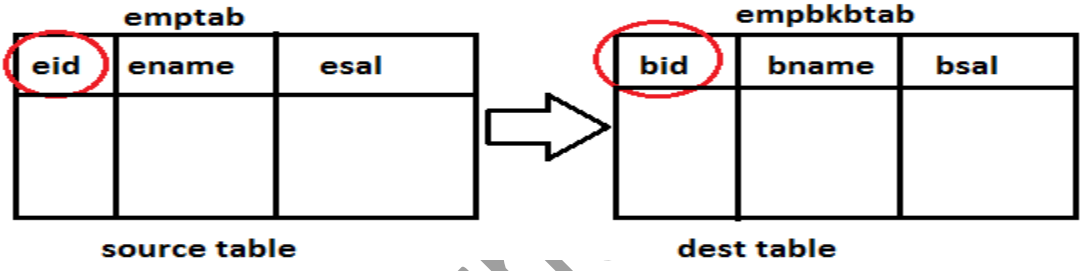
javabyraghu@gmail.com

I

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com



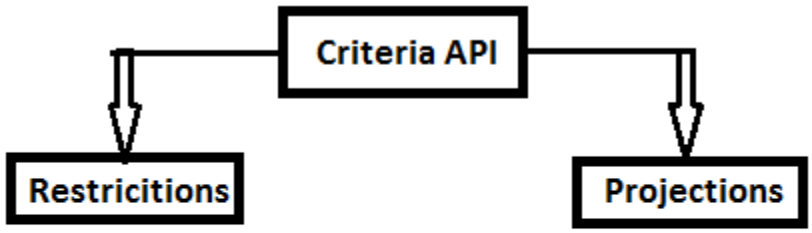
javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

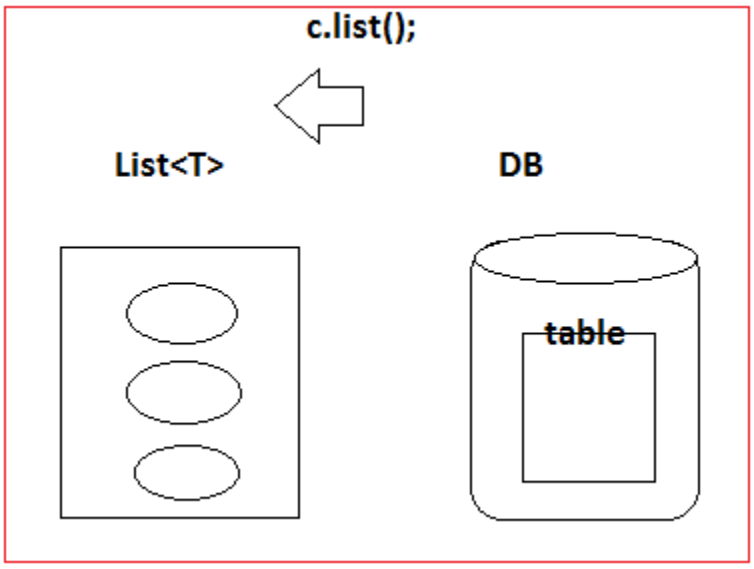

javabyraghu@gmai



[where...]
To specify Row details

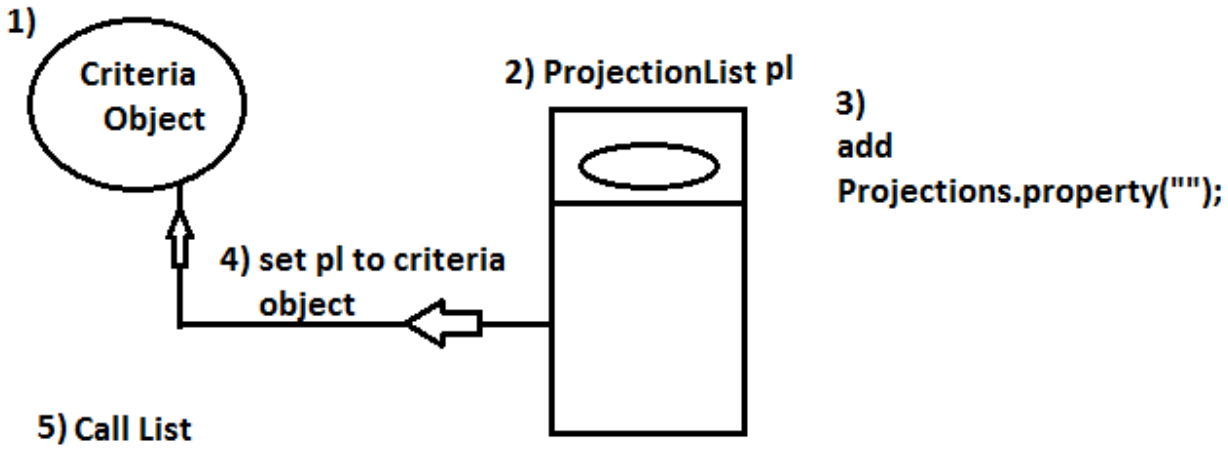
[select...]
To specify Column details

javabyraghu@gmail.com



javabyraghu@gmail.com

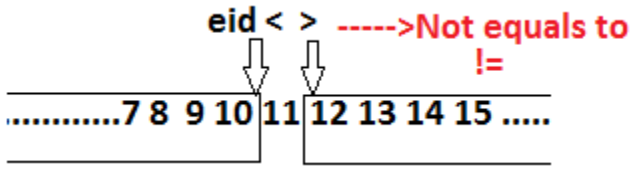
javabyraghu@gmail.com



javabyraghu@gm

javabyraghu@gmail.com

javabyraghu@gmail.com

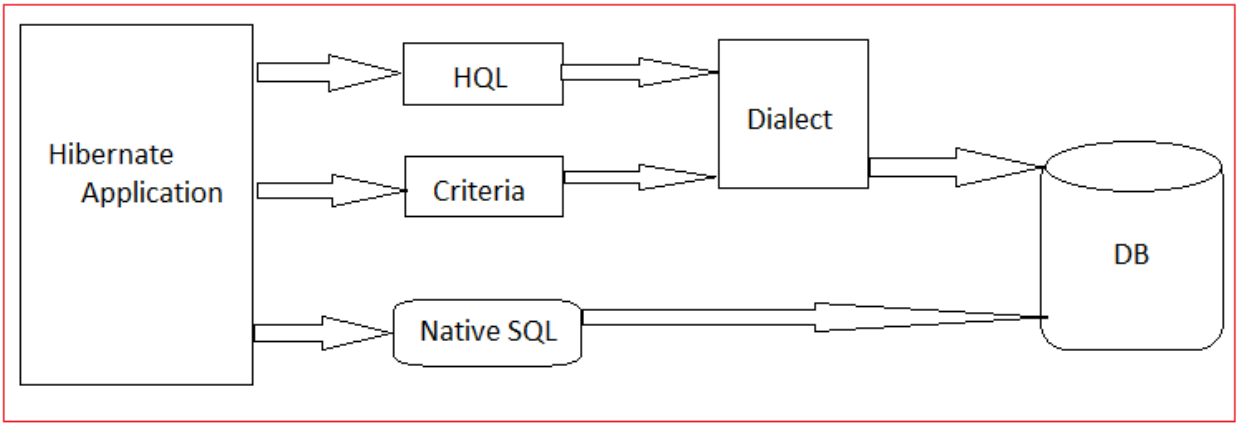


javabyraghu@gmail.com

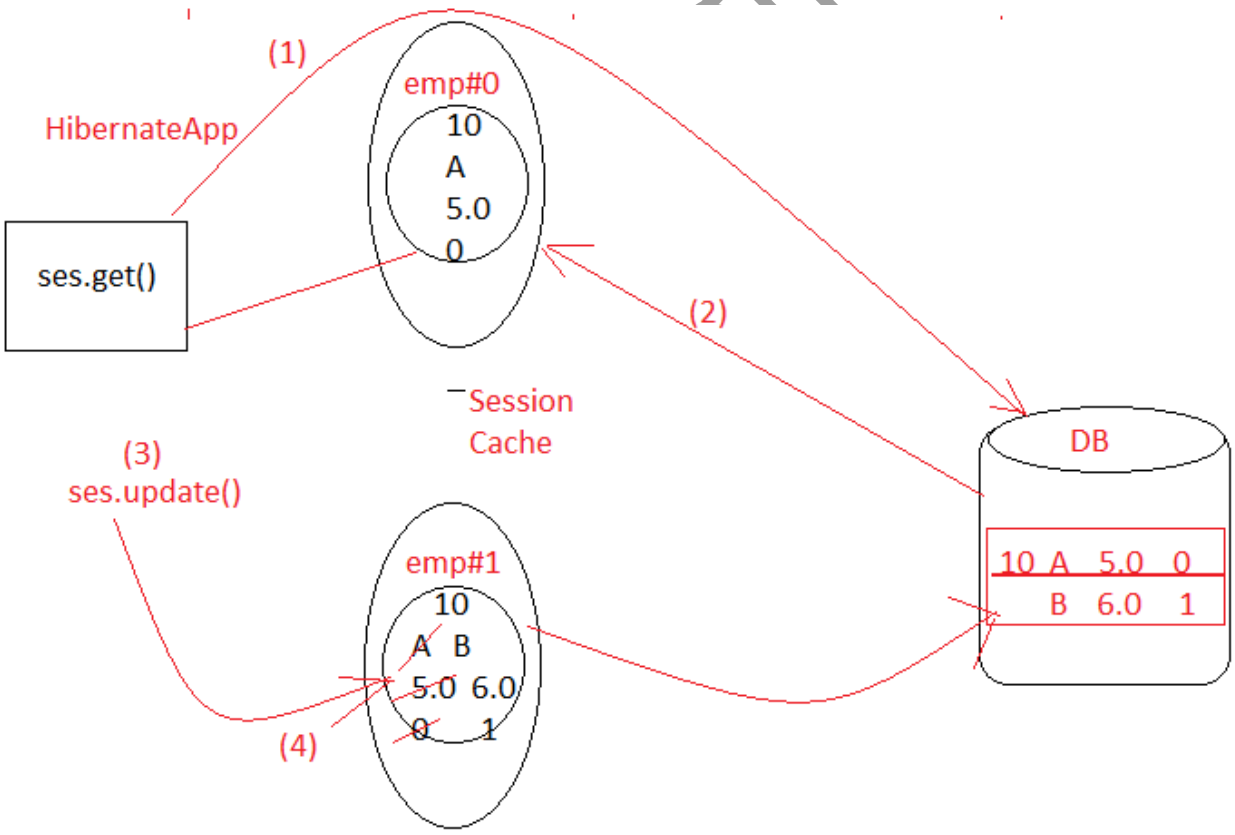
javabyraghu@gmail.com

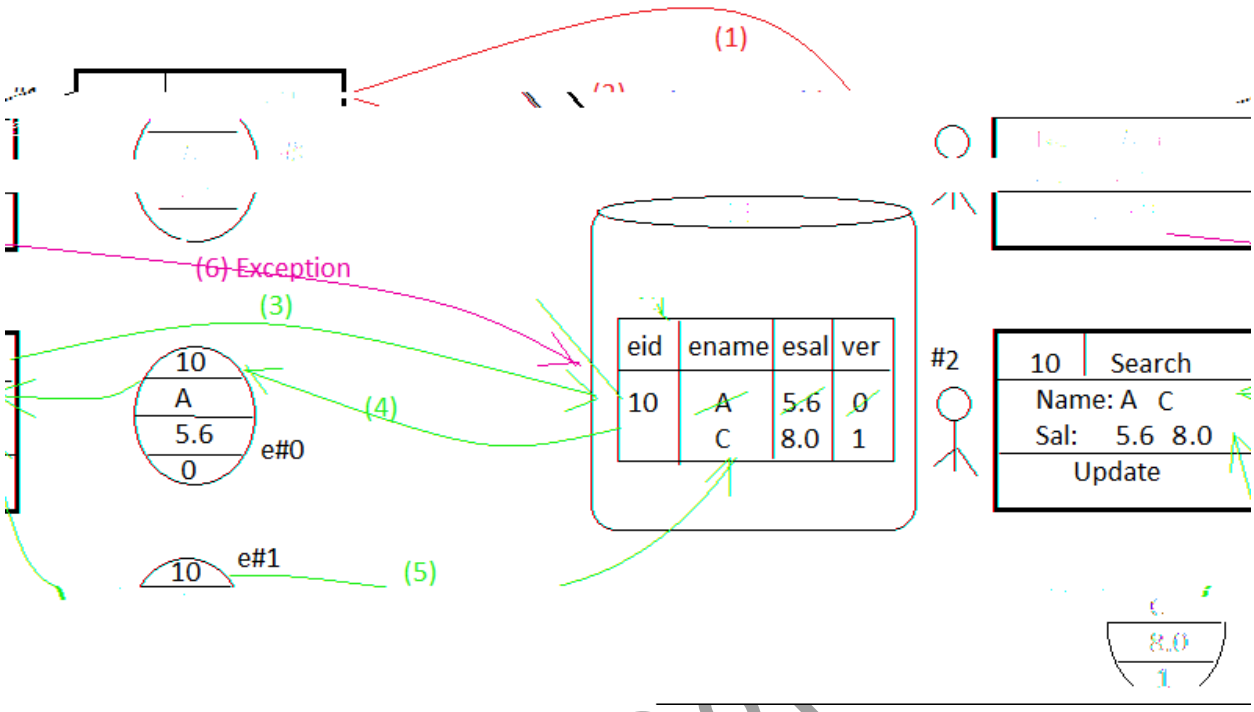
javabyraghu@gmail.com

javabyraghu@gmail.com



javabyraghu.com


(1)
HibernateApp
emp#0
10
A
5.0
0
ses.get()
(2)
Session
Cache
DB
10 A 5.0 0
B 6.0 1
(3)
ses.update()
emp#1
10
A B
5.0 6.0
0 1
(4)



javabyraghu@g

javabyraghu@gmail.com

javabyraghu@gm

javabyraghu@gmail.com

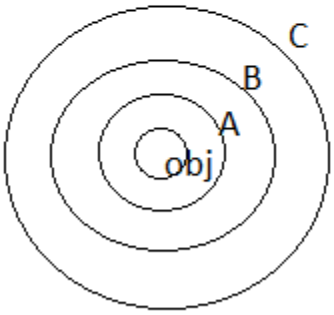
javabyraghu@gmail.com

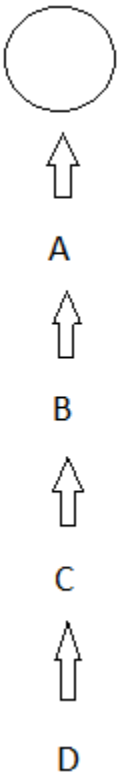
javabyraghu@gmail.com



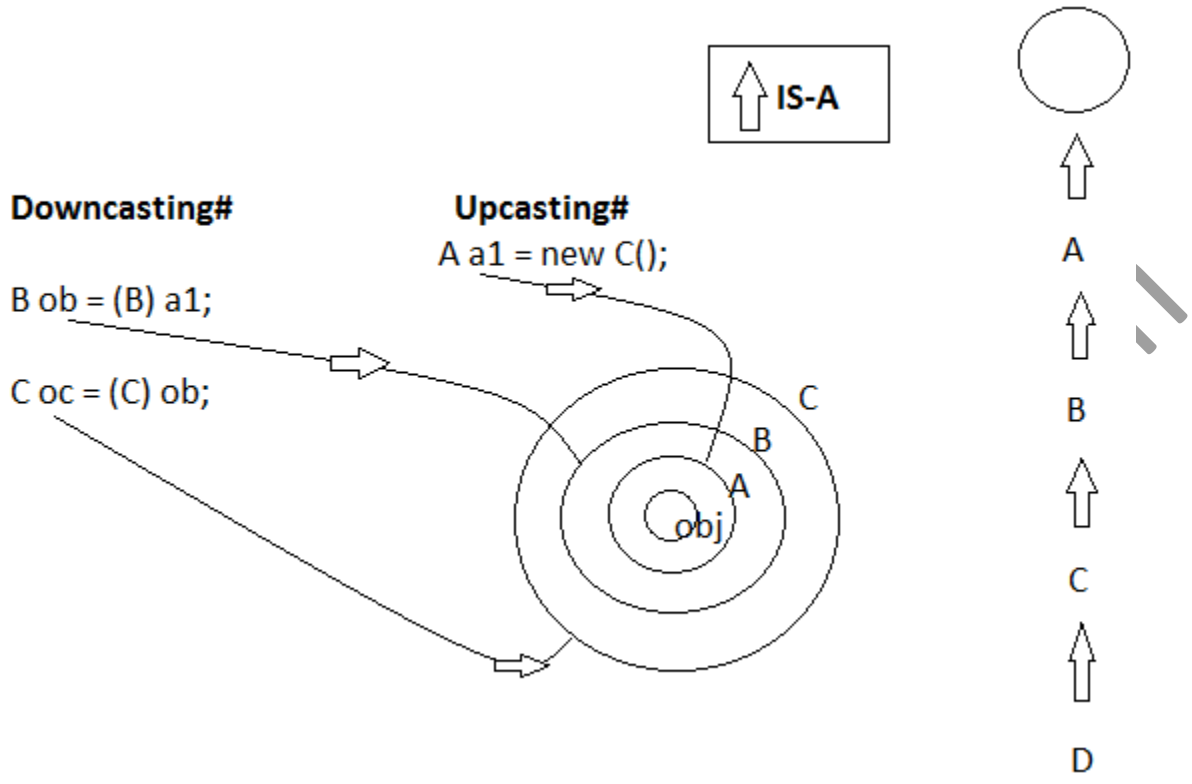
Upcasting#

A a1 = new C();





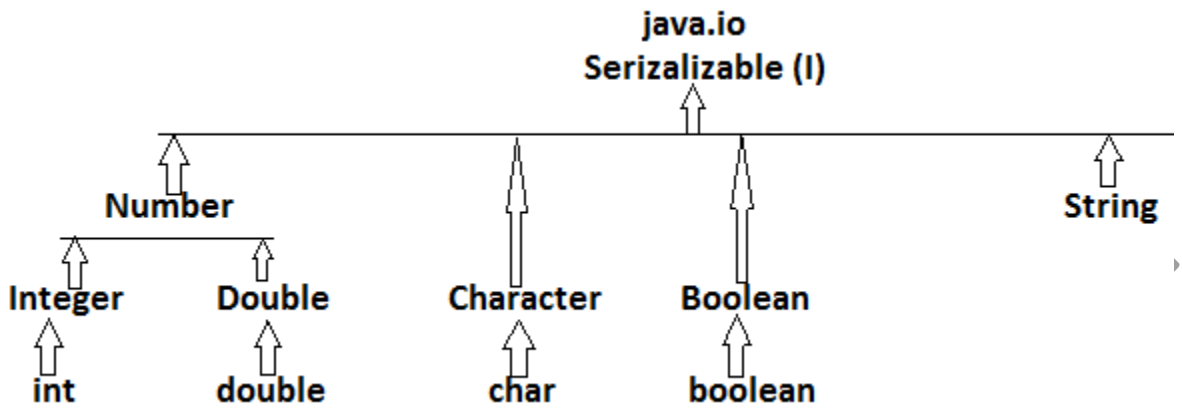
javabyraghu@gmail.com



→

I

javabyraghu

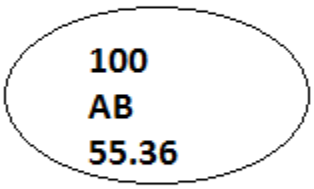


→

Model Class

Employee(empId,empName,empSal)

int

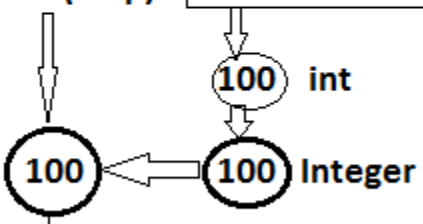


DB Table

emptab

eid	ename	esal
100	AB	55.36

ses.save(emp)



Code:

Serializable s = ses.save(emp);

Serializable

javabyraghu@gmail.com

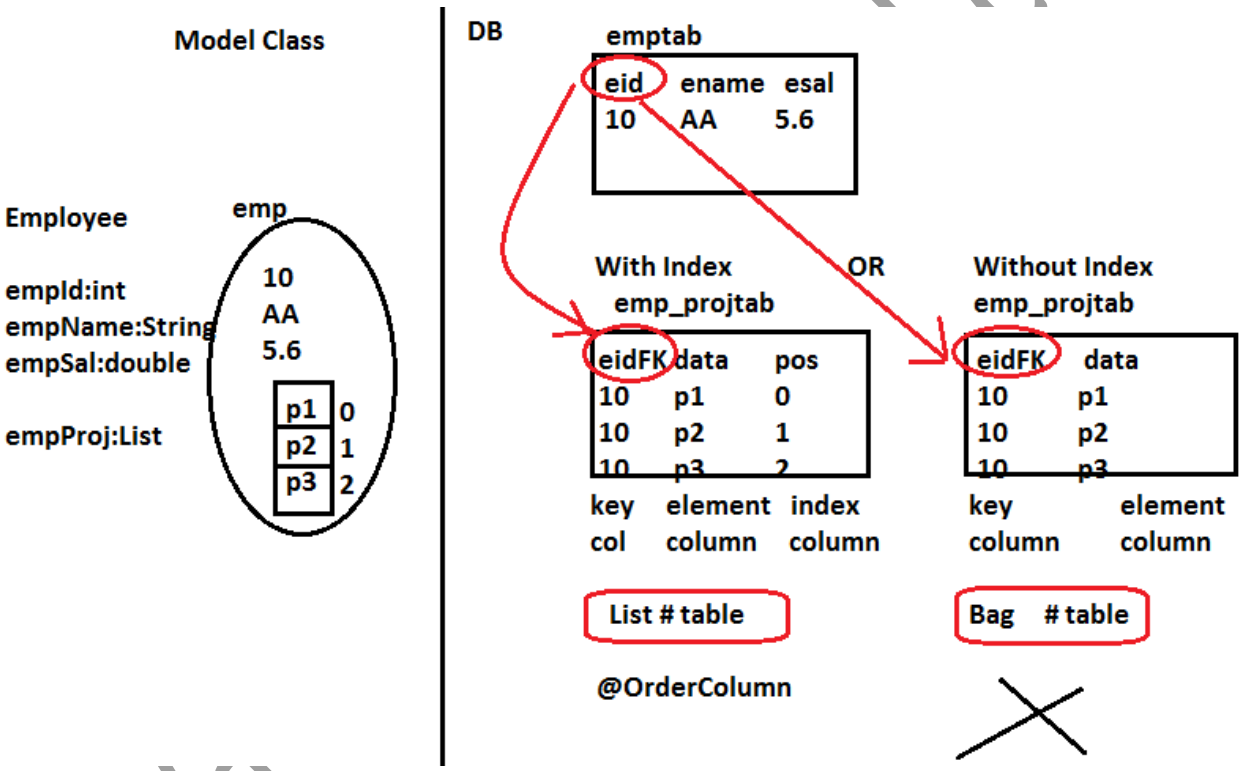
javabyraghu@gmail.com

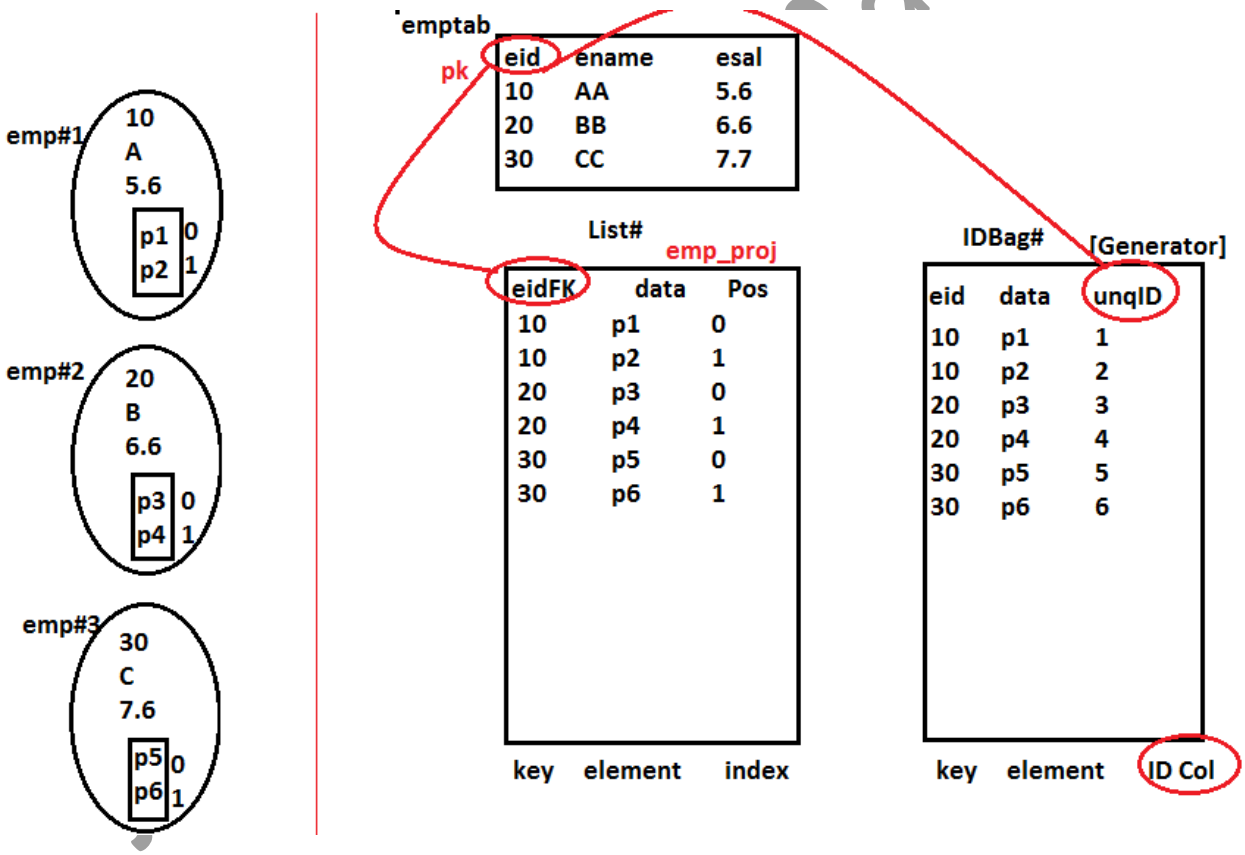
javabyraghu@gmail.

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com


Model Class
DB
emptab
eid ename esal
10 AA 5.6
Employee
emp
empId:int
empName:String
empSal:double
empProj:List
10
AA
5.6
p1 0
p2 1
p3 2
With Index
emp_projtab
OR
Without Index
emp_projtab
eidFK data pos
10 p1 0
10 p2 1
10 p3 2
eidFK data
10 p1
10 p2
10 p3
key element index
col column column
key element
column column
List # table
Bag # table
@OrderColumn


emp#1
10
A
5.6
p1 0
p2 1
emp#2
20
B
6.6
p3 0
p4 1
emp#3
30
C
7.6
p5 0
p6 1
emptab
pk
eid ename esal
10 AA 5.6
20 BB 6.6
30 CC 7.7
List#
emp_proj
eidFK data Pos
10 p1 0
10 p2 1
20 p3 0
20 p4 1
30 p5 0
30 p6 1
key element index
IDBag#
[Generator]
eid data unqID
10 p1 1
10 p2 2
20 p3 3
20 p4 4
30 p5 5
30 p6 6
key element ID Col

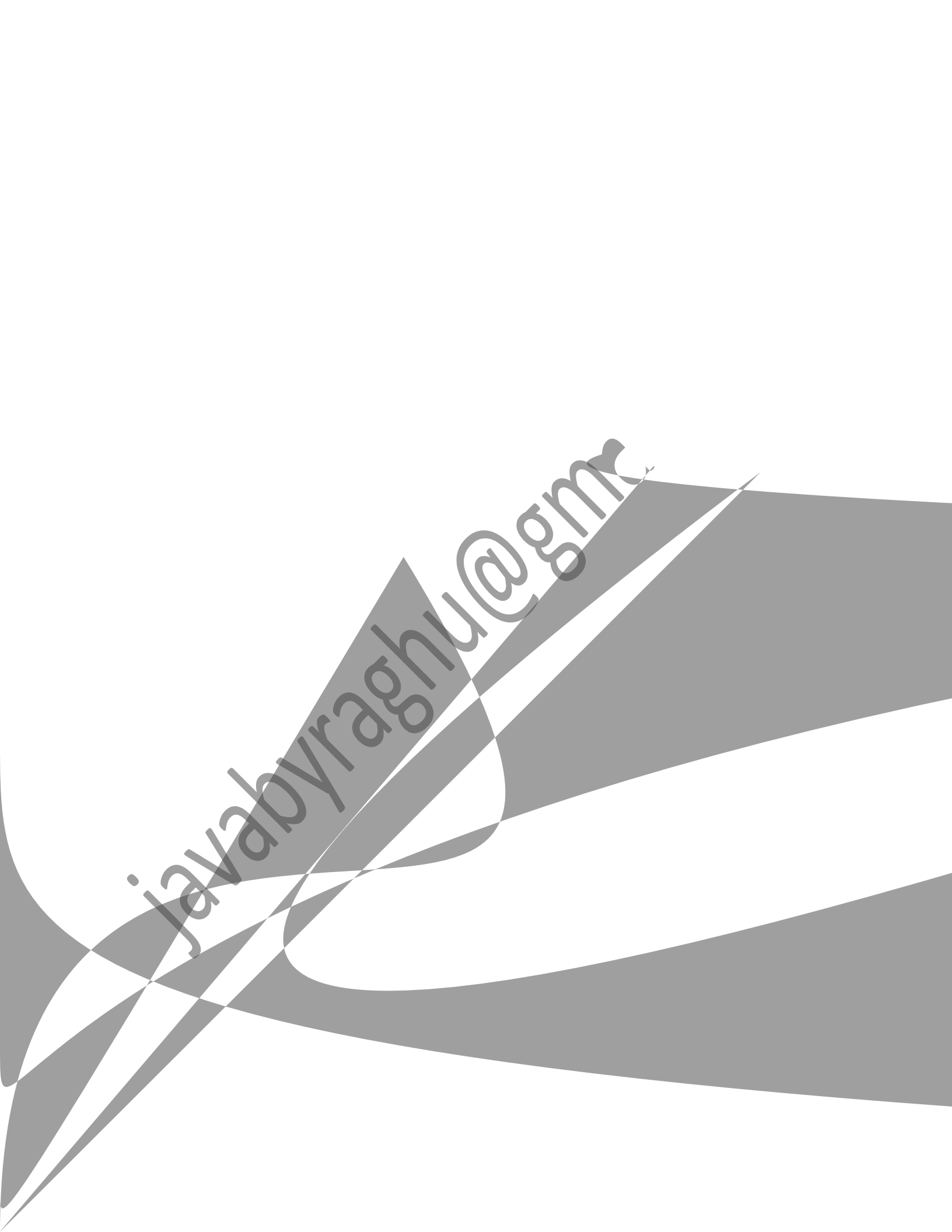
javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com


javabyraghu@gm

javabyraghu@gmail.com

javabyraghu@gmail.com

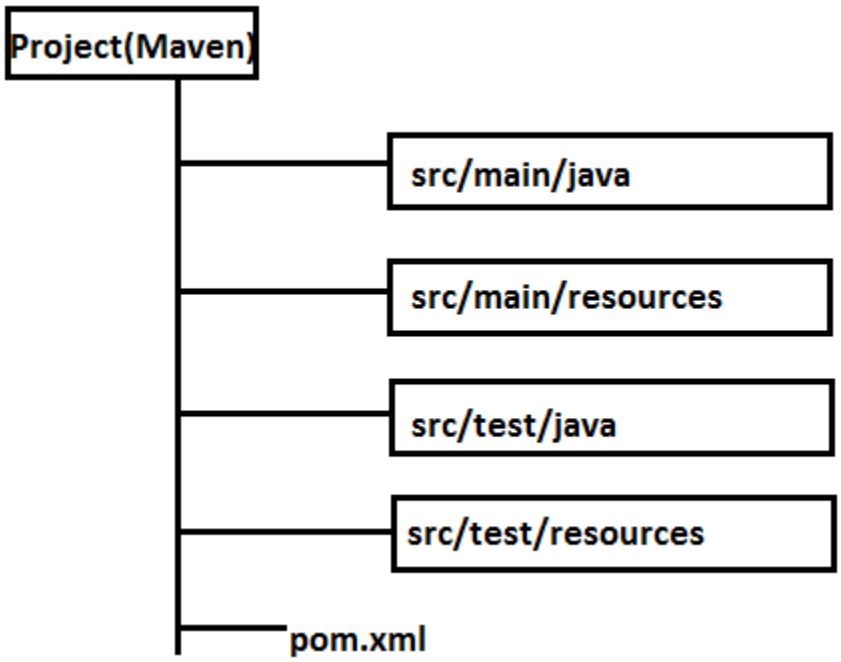
javabyraghu@gmail.com

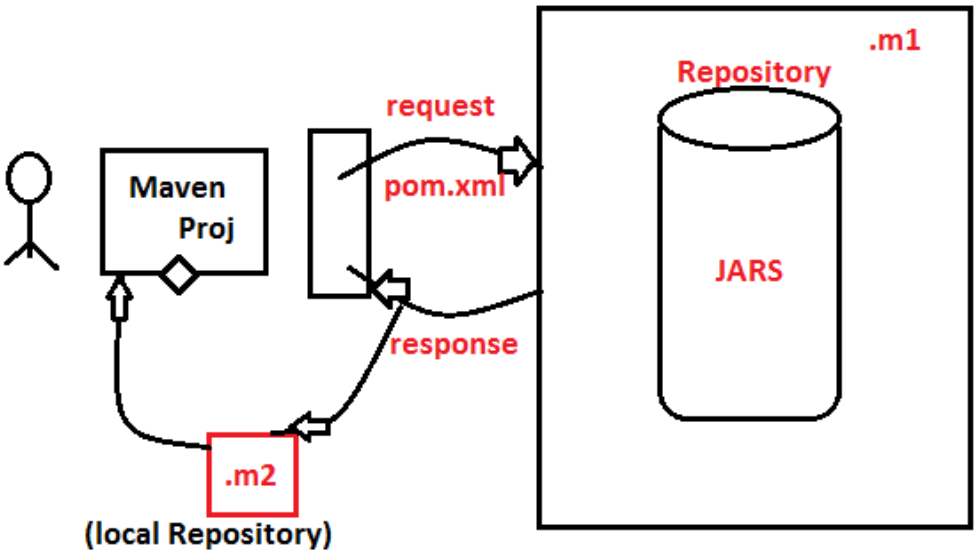
javabyraghu@gmail.com

javabyraghu@gmail.com

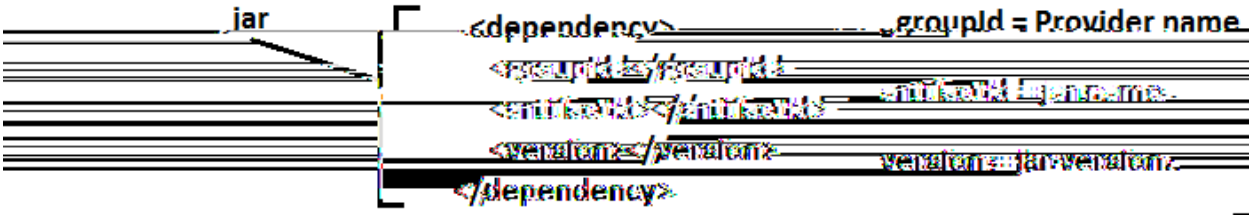
javabyraghu@gmail.com

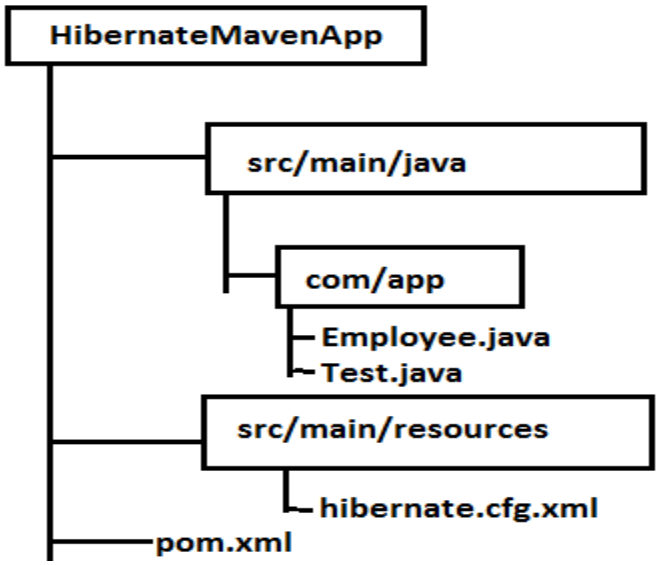
javabyraghu@gmail.com


Project(Maven)
src/main/java
src/main/resources
src/test/java
src/test/resources
pom.xml


.m1
Repository
request
pom.xml
JARS
Maven
Proj
response
.m2
(local Repository)

pom.xml

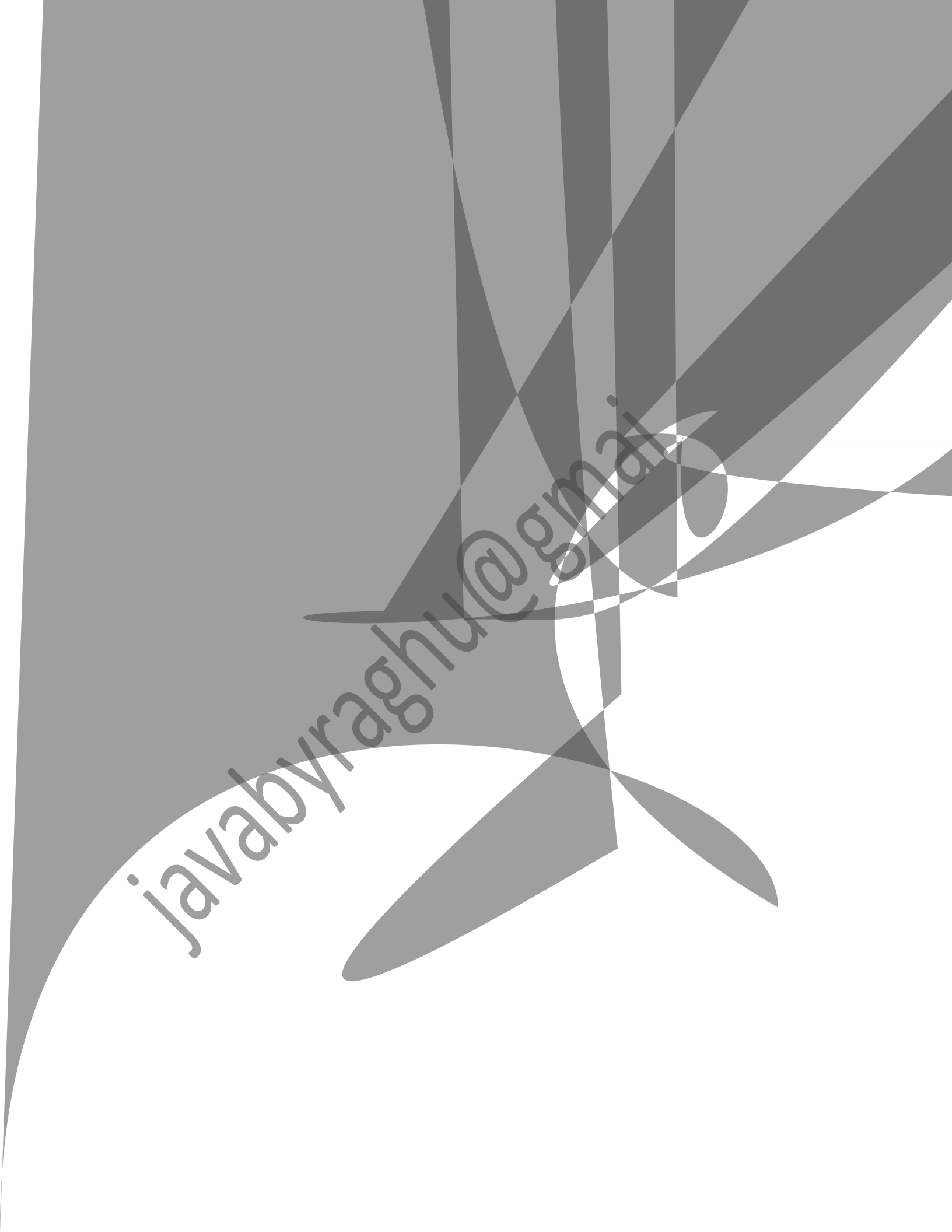

jar
<dependency>
groupId = Provider name
</dependency>



javabyraghu@gmail.com

javabyraghu@gmail.com

javabyraghu@gmail.com


javabyraghu@gmai

javabyraghu@gmail.co

javabyraghu@gmail.co


javabyraghu@gmail.c

I

javabyraghu@gmail.com

→

I

→

I

javabyraghu@gmail.com